

4-2-2a

t (seconds)	0	8	20	25	32	40
$v(t)$ (meters per second)	3	5	-10	-8	-4	7

The velocity of a particle moving along the x -axis is modeled by a differentiable function v , where the position x is measured in meters, and time t is measured in seconds. Selected values of $v(t)$ are given in the table above. The particle is at position $x=7$ meters when $t=0$ seconds.

1. Estimate the acceleration of the particle at $t=36$ seconds. Show the computations that lead to your answer. Indicate units of measure.

Part A

1 point is earned for the answer

$$a(36) = v'(36) \approx \frac{v(40) - v(32)}{40 - 32} = \frac{11}{8} \text{ meters/sec}^2$$



0	1
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The student response earns one of the following points:

1 point is earned for the answer

$$a(36) = v'(36) \approx \frac{v(40) - v(32)}{40 - 32} = \frac{11}{8} \text{ meters/sec}^2$$

4-2-2a

2. Suppose that the acceleration of the particle is positive for $0 < t < 8$ seconds. Explain why the position of the particle at $t=8$ seconds must be greater than $x=30$ meters.

Part D

1 point is earned for $v'(t)=a(t)$

1 point is earned for the explanation of $x(8) > 30$

Since $v'(t)=a(t) > 0$ for $0 < t < 8$, $v(t) \geq 3$ on this interval.

Therefore, $x(8)=x(0)+\int_0^8 v(t)dt \geq 7 + 8.3 > 30$.



0	1	2
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The student response earns all of the following points:

1 point is earned for $v'(t)=a(t)$

1 point is earned for the explanation of $x(8) > 30$

Since $v'(t)=a(t) > 0$ for $0 < t < 8$, $v(t) \geq 3$ on this interval.

Therefore, $x(8)=x(0)+\int_0^8 v(t)dt \geq 7 + 8.3 > 30$.

4-2-2a

3. For $0 \leq t \leq 40$, must the particle change direction in any of the subintervals indicated by the data in the table? If so, identify the subintervals and explain your reasoning. If not, explain why not.
-

Part C

1 point is earned for the answer

1 point is earned for the explanation

$$v(8) > 0 \text{ and } v(20) < 0$$

$$v(32) < 0 \text{ and } v(40) > 0$$

Therefore, the particle changes direction in the intervals $8 < t < 20$ and $32 < t < 40$.



0	1	2
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The student response earns all of the following points:

1 point is earned for the answer

1 point is earned for the explanation

$$v(8) > 0 \text{ and } v(20) < 0$$

$$v(32) < 0 \text{ and } v(40) > 0$$

Therefore, the particle changes direction in the intervals $8 < t < 20$ and $32 < t < 40$.

4-2-2a

4.

t (seconds)	0	8	20	25	32	40
$v(t)$ (meters per second)	3	5	-10	-8	-4	7

The velocity of a particle moving along the x -axis is modeled by a differentiable function v , where the position x is measured in meters, and time t is measured in seconds. Selected values of $v(t)$ are given in the table above. The particle is at position $x = 7$ meters when $t = 0$ seconds.

(a) Estimate the acceleration of the particle at $t = 36$ seconds. Show the computations that lead to your answer. Indicate units of measure.

(b) Using correct units, explain the meaning of $\int_{20}^{40} v(t) dt$ in the context of this problem. Use a trapezoidal sum with the three subintervals indicated by the data in the table to approximate $\int_{20}^{40} v(t) dt$.

(c) For $0 \leq t \leq 40$, must the particle change direction in any of the subintervals indicated by the data in the table? If so, identify the subintervals and explain your reasoning. If not, explain why not.

(d) Suppose that the acceleration of the particle is positive for $0 < t < 8$ seconds. Explain why the position of the particle at $t = 8$ seconds must be greater than $x = 30$ meters.

Part A

1 point is earned for the answer

$$a(36) = v'(36) \approx \frac{v(40) - v(32)}{40 - 32} = \frac{11}{8} \text{ meters/sec}^2$$

1 point is earned for units in (a) and (b)



0	1	2
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The student response earns all of the following points:

1 point is earned for the answer

$$a(36) = v'(36) \approx \frac{v(40) - v(32)}{40 - 32} = \frac{11}{8} \text{ meters/sec}^2$$

1 point is earned for units in (a) and (b)

4-2-2a

Part B

1 point is earned for meaning of $\int_{20}^{40} v(t) dt$

2 points are earned for the trapezoidal approximation

$\int_{20}^{40} v(t) dt$ is the particle's change in position in meters from time $t = 20$ seconds to time $t = 40$ seconds.

$$\int_{20}^{40} v(t) dt \approx \frac{v(20)+v(25)}{2} \cdot 5 + \frac{v(25)+v(32)}{2} \cdot 7 + \frac{v(32)+v(40)}{2} \cdot 8$$

$$= -75 \text{ meters}$$

1 point is earned for units in (a) and (b)



0	1	2	3	4
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The student response earns all of the following points:

1 point is earned for meaning of $\int_{20}^{40} v(t) dt$

2 points are earned for the trapezoidal approximation

$\int_{20}^{40} v(t) dt$ is the particle's change in position in meters from time $t = 20$ seconds to time $t = 40$ seconds.

$$\int_{20}^{40} v(t) dt \approx \frac{v(20)+v(25)}{2} \cdot 5 + \frac{v(25)+v(32)}{2} \cdot 7 + \frac{v(32)+v(40)}{2} \cdot 8$$

$$= -75 \text{ meters}$$

1 point is earned for units in (a) and (b)

Part C

1 point is earned for the answer

1 point is earned for the explanation

$$v(8) > 0 \text{ and } v(20) < 0$$

$$v(32) < 0 \text{ and } v(40) > 0$$

Therefore, the particle changes direction in the intervals $8 < t < 20$ and $32 < t < 40$.

4-2-2a



0	1	2
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The student response earns all of the following points:

1 point is earned for the answer

1 point is earned for the explanation

$$v(8) > 0 \text{ and } v(20) < 0$$

$$v(32) < 0 \text{ and } v(40) > 0$$

Therefore, the particle changes direction in the intervals $8 < t < 20$ and $32 < t < 40$.

Part D

1 point is earned for $v'(t) = a(t)$

1 point is earned for the explanation of $x(8) > 30$

Since $v'(t) = a(t) > 0$ for $0 < t < 8$, $v(t) \geq 3$ on this interval.

$$\text{Therefore, } x(8) = x(0) + \int_0^8 v(t) dt \geq 7 + 8 \cdot 3 > 30.$$



0	1	2
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The student response earns all of the following points:

1 point is earned for $v'(t) = a(t)$

1 point is earned for the explanation of $x(8) > 30$

Since $v'(t) = a(t) > 0$ for $0 < t < 8$, $v(t) \geq 3$ on this interval.

$$\text{Therefore, } x(8) = x(0) + \int_0^8 v(t) dt \geq 7 + 8 \cdot 3 > 30.$$

4-2-2a

5.  The velocity of a particle moving along the x -axis is given by $v(t) = \sqrt{t} - \cos(e^t)$ for $t \geq 0$. Which of the following statements describes the motion of the particle at $t = 1$?
- (A) The particle is moving to the left with positive acceleration.
 - (B) The particle is moving to the right with positive acceleration. ✓
 - (C) The particle is moving to the left with negative acceleration.
 - (D) The particle is moving to the right with negative acceleration.

Answer B

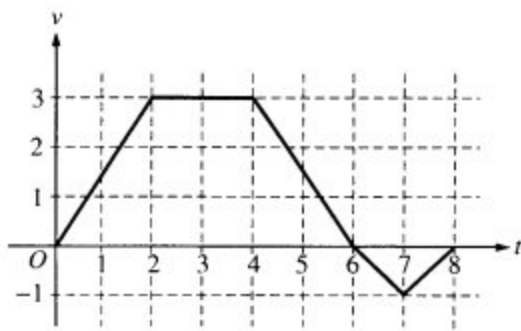
This option is correct. At $t = 1$, the graph of v is positive. In addition, the graph of v is increasing at $t = 1$, which means that $v'(1)$ is positive. Alternatively, $v(1) = 1.912$ and $v'(1) = 1.617$.

Since velocity is the derivative of position, a positive value of v at $t = 1$ means that x is increasing, i.e. the particle is moving to the right. Since acceleration is the derivative of velocity, a positive value of v' at $t = 1$ means that the particle is moving with positive acceleration.

6. The position of a particle moving along the x -axis is given by a twice-differentiable function $x(t)$. If $x(2) < 0$, $x'(2) < 0$, and $x''(2) < 0$, which of the following statements must be true about the particle at time $t = 2$?
- (A) The particle is moving toward the origin at a decreasing speed.
 - (B) The particle is moving toward the origin at an increasing speed.
 - (C) The particle is moving away from the origin at a decreasing speed.
 - (D) The particle is moving away from the origin at an increasing speed. ✓
 - (E) The particle is moving away from the origin at a constant speed.

4-2-2a

7.



A bug begins to crawl up a vertical wire at time $t = 0$. The velocity v of the bug at time t , $0 \leq t \leq 8$, is given by the function whose graph is shown above.

At what value of t does the bug change direction?

- (A) 2
- (B) 4
- (C) 6
- (D) 7
- (E) 8



8.  A particle moves along a line so that at time t , where $0 \leq t \leq \pi$, its position is given by $s(t) = -4 \cos t - \frac{t^2}{2} + 10$. What is the velocity of the particle when its acceleration is zero?

- (A) -5.19
- (B) 0.74
- (C) 1.32
- (D) 2.55
- (E) 8.13



9.  A particle moves along the x -axis so that at any time $t \geq 0$, its velocity is given by $v(t) = 3 + 4.1 \cos(0.9t)$. What is the acceleration of the particle at time $t = 4$?

- (A) -2.016
- (B) -0.677
- (C) 1.633
- (D) 1.814
- (E) 2.978



4-2-2a

10. A particle moves along the x -axis so that its position at time t is given by $x(t) = t^2 - 6t + 5$. For what value of t is the velocity of the particle zero?
- (A) 1
(B) 2
(C) 3 ✓
(D) 4
(E) 5
11.  A particle moves along a straight line with velocity given by $v(t) = 5 + e^{t/3}$ for time $t \geq 0$. What is the acceleration of the particle at time $t = 4$?
- (A) 0.422
(B) 0.698
(C) 1.265 ✓
(D) 8.794
(E) 28.381
12.  A particle moves along the x -axis so that its position at time $t > 0$ is given by $x(t)$ and $\frac{dx}{dt} = -10t^4 + 9t^2 + 8t$. The acceleration of the particle is zero when $t =$
- (A) 0.387
(B) 0.831 ✓
(C) 1.243
(D) 1.647
(E) 8.094
13.  A particle moves along the x -axis so that at time $t \geq 0$ the position of the particle is given by $x(t) = 0.5t^4 - 1.5t^3 - 2t^2 + 6t - 1$. What is the velocity of the particle at the first instance the particle is at the origin?
- (A) -4.071
(B) -2.048
(C) 0
(D) 5.153 ✓
(E) 6

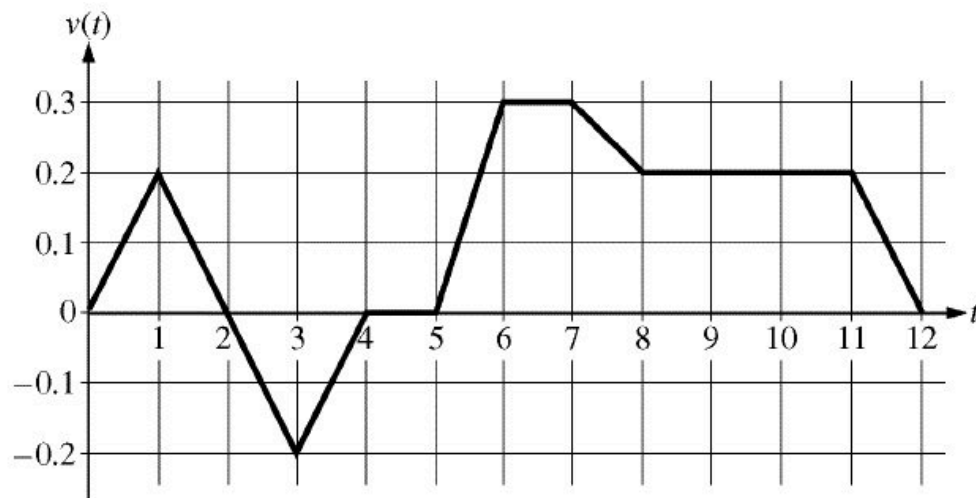
4-2-2a

14. The position of a particle moving along a straight line at any time t is given by $s(t) = t^2 + 4t + 4$. What is the acceleration of the particle when $t = 4$?
- (A) 0
- (B) 2
- (C) 4
- (D) 8
- (E) 12

Answer B

15.  Two particles start at the origin and move along the x -axis. For $0 \leq t \leq 10$, their respective position functions are given by $x_1 = \sin t$ and $x_2 = e^{-2t} - 1$. For how many values of t do the particles have the same velocity?
- (A) None
- (B) One
- (C) Two
- (D) Three
- (E) Four
16.  For $t \geq 0$, the velocity of a particle moving along the x -axis is given by $v(t) = t^3 - 6t^2 + 10t - 4$. At what time t does the direction of motion of the particle change from right to left?
- (A) 0.586
- (B) 1.184
- (C) 2.000
- (D) 2.816

4-2-2a



Caren rides her bicycle along a straight road from home to school, starting at home at time $t=0$ minutes and arriving at school at time $t=12$ minutes. During the time interval $0 \leq t \leq 12$ minutes, her velocity $v(t)$, in miles per minute, is modeled by the piecewise-linear function whose graph is shown above.

17. Shortly after leaving home, Caren realizes she left her calculus homework at home, and she returns to get it. At what time does she turn around to go back home? Give a reason for your answer.

Part C

1 point is earned for answer

1 point is earned for reason

Caren turns around to go back home at time $t=2$ minutes. This is the time at which her velocity changes from positive to negative.



0	1	2
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The student response earns two of the following points:

1 point is earned for answer

1 point is earned for reason

Caren turns around to go back home at time $t=2$ minutes. This is the time at which her velocity changes from positive to negative.

4-2-2a

18. Find the acceleration of Caren's bicycle at time $t=7.5$ minutes. Indicate units of measure.
-

Part A

1 point is earned for answer

1 point is earned for unit

$$a(7.5) = v'(7.5) = \frac{v(8)-v(7)}{8-7} = -0.1 \text{ miles /minute}^2$$



0	1	2
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The student response earns two of the following points:

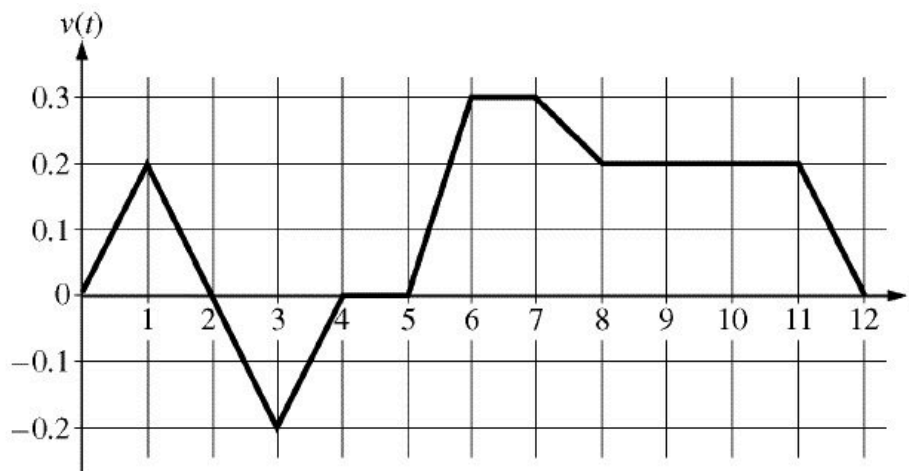
1 point is earned for answer

1 point is earned for unit

$$a(7.5) = v'(7.5) = \frac{v(8)-v(7)}{8-7} = -0.1 \text{ miles /minute}^2$$

4-2-2a

19.



Caren rides her bicycle along a straight road from home to school, starting at home at time $t = 0$ minutes and arriving at school at time $t = 12$ minutes. During the time interval $0 \leq t \leq 12$ minutes, her velocity $v(t)$, in miles per minute, is modeled by the piecewise-linear function whose graph is shown above.

(a) Find the acceleration of Caren's bicycle at time $t = 7.5$ minutes. Indicate units of measure.

(b) Using correct units, explain the meaning of $\int_0^{12} |v(t)| dt$ in terms of Caren's trip. Find the value of $\int_0^{12} |v(t)| dt$.

(c) Shortly after leaving home, Caren realizes she left her calculus homework at home, and she returns to get it. At what time does she turn around to go back home? Give a reason for your answer.

(d) Larry also rides his bicycle along a straight road from home to school in 12 minutes. His velocity is modeled by the function w given by $w(t) = \frac{\pi}{15} \sin\left(\frac{\pi}{12}t\right)$, where $w(t)$ is in miles per minute for $0 \leq t \leq 12$ minutes. Who lives closer to school: Caren or Larry? Show the work that leads to your answer.

Part A

1 point is earned for the answer

1 point is earned for units

$$a(7.5) = v'(7.5) = \frac{v(8) - v(7)}{8 - 7} = -0.1 \text{ miles /minute}^2$$



0	1	2
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The student response earns all of the following points:

1 point is earned for the answer

4-2-2a

1 point is earned for units

$$a(7.5) = v'(7.5) = \frac{v(8) - v(7)}{8 - 7} = -0.1 \text{ miles / minute}^2$$

Part B

1 point is earned for the meaning of integral

1 point is earned for the value of integral

$\int_0^{12} |v(t)| dt$ is the total distance, in miles, that Caren rode during the 12 minutes from $t = 0$ to $t = 12$.

$$\int_0^{12} |v(t)| dt = \int_0^2 v(t) dt - \int_2^4 v(t) dt + \int_4^{12} v(t) dt$$

$$= 0.2 + 0.2 + 1.4 = 1.8 \text{ miles}$$



0	1	2
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The student response earns all of the following points:

1 point is earned for the meaning of integral

1 point is earned for the value of integral

$\int_0^{12} |v(t)| dt$ is the total distance, in miles, that Caren rode during the 12 minutes from $t = 0$ to $t = 12$.

$$\int_0^{12} |v(t)| dt = \int_0^2 v(t) dt - \int_2^4 v(t) dt + \int_4^{12} v(t) dt$$

$$= 0.2 + 0.2 + 1.4 = 1.8 \text{ miles}$$

Part C

1 point is earned for the answer

4-2-2a

1 point is earned for reason

Caren turns around to go back home at time $t = 2$ minutes. This is the time at which her velocity changes from positive to negative.



0	1	2
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The student response earns all of the following points:

1 point is earned for the answer

1 point is earned for reason

Caren turns around to go back home at time $t = 2$ minutes. This is the time at which her velocity changes from positive to negative..

Part D

2 points are earned for Larry's distance from school where 1 point is earned for integral and 1 point is earned for value

1 point is earned for Caren's distance from school and conclusion

$$\int_0^{12} w(t) dt = 1.6; \text{ Larry lives 1.6 miles from school.}$$

$$\int_0^{12} v(t) dt = 1.4; \text{ Caren lives 1.4 miles from school.}$$

Therefore, Caren lives closer to school.



0	1	2	3
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The students response earns all of the following points:

2 points are earned for Larry's distance from school where 1 point is earned for integral and 1 point is earned for value

1 point is earned for Caren's distance from school and conclusion

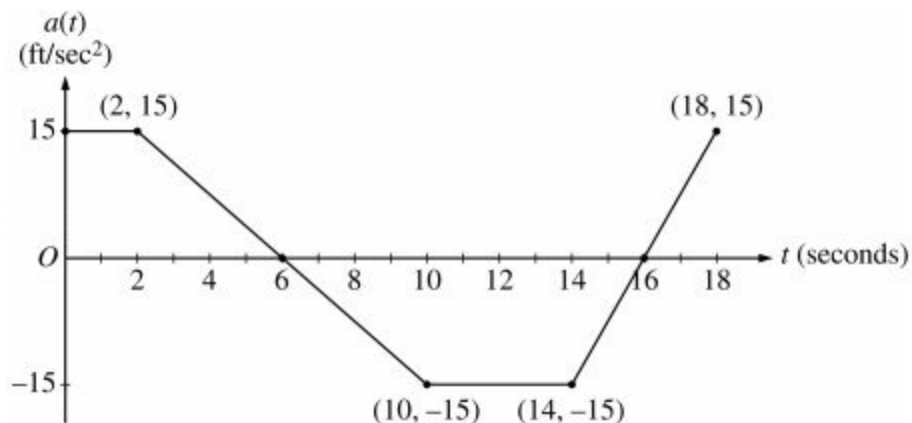
$$\int_0^{12} w(t) dt = 1.6; \text{ Larry lives 1.6 miles from school.}$$

4-2-2a

$$\int_0^{12} v(t) dt = 1.4; \text{ Caren lives 1.4 miles from school.}$$

Therefore, Caren lives closer to school.

20.



A car is traveling on a straight road with velocity 55 ft/sec at time $t = 0$. For $0 \leq t \leq 18$ seconds, the car's acceleration $a(t)$, in ft/sec^2 , is the piecewise linear function defined by the graph above

Is the velocity of the car increasing at $t = 2$ seconds? Why or why not?

Part A

1 point is earned for the answer and reason

Since $v'(2) = a(2)$ and $a(2) = 15 > 0$, the velocity is increasing at $t = 2$.



0

1

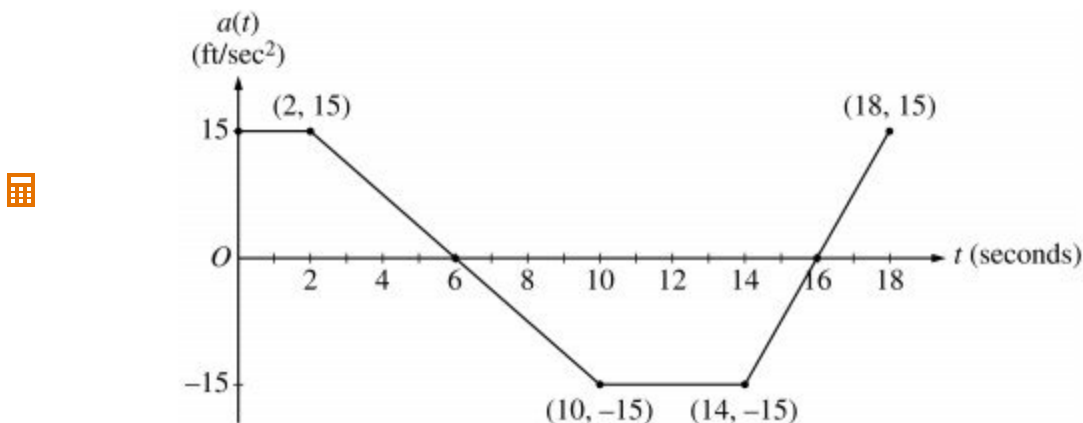
The student response earns all of the following points:

1 point is earned for the answer and reason

Since $v'(2) = a(2)$ and $a(2) = 15 > 0$, the velocity is increasing at $t = 2$.

4-2-2a

21.



A car is traveling on a straight road with velocity 55 ft/sec at time $t = 0$. For $0 \leq t \leq 18$ seconds, the car's acceleration $a(t)$, in ft/sec², is the piecewise linear function defined by the graph above.

- (a) Is the velocity of the car increasing at $t = 2$ seconds? Why or why not?
- (b) At what time in the interval $0 \leq t \leq 18$, other than $t = 0$, is the velocity of the car 55 ft/sec? Why?
- (c) On the time interval $0 \leq t \leq 18$, what is the car's absolute maximum velocity, in ft/sec, and at what time does it occur? Justify your answer.
- (d) At what times in the interval $0 \leq t \leq 18$, if any, is the car's velocity equal to zero? Justify your answer.

Part A

1 point is earned for the answer and reason

Since $v'(2) = a(2)$ and $a(2) = 15 > 0$, the velocity is increasing at $t = 2$.



0	1
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The student response earns all of the following points:

1 point is earned for the answer and reason

Since $v'(2) = a(2)$ and $a(2) = 15 > 0$, the velocity is increasing at $t = 2$.

Part B

1 point is earned for: $t = 12$

1 point is earned for the reason

At time $t = 12$ because $v(12) - v(0) = \int_0^{12} a(t) dt = 0$.

4-2-2a



0	1	2
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The student response earns all of the following points:

1 point is earned for: $t = 12$

1 point is earned for the reason

At time $t = 12$ because $v(12) - v(0) = \int_0^{12} a(t) dt = 0$.

Part C

1 point is earned for: $t = 6$

1 point is earned for the absolute maximum velocity

1 point is earned for identifying $t = 6$ and $t = 18$ as candidates or indicating that v increases, decreases, then increases.

1 point is earned for eliminating $t = 18$

The absolute maximum velocity is 115 ft/sec at $t = 6$.

The absolute maximum must occur at $t = 6$ or at an endpoint.

$$\begin{aligned}
 v(6) &= 55 + \int_0^6 a(t) dt \\
 &= 55 + 2(15) + \frac{1}{2}(4)(15) = 115 > v(0) \\
 \int_6^{18} a(t) dt &< 0 \text{ so } v(18) < v(6)
 \end{aligned}$$



0	1	2	3	4
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The student response earns all of the following points:

1 point is earned for: $t = 6$

1 point is earned for the absolute maximum velocity

4-2-2a

1 point is earned for identifying $t = 6$ and $t = 18$ as candidates or indicating that v increases, decreases, then increases.

1 point is earned for eliminating $t = 18$

The absolute maximum velocity is 115 ft/sec at $t = 6$.

The absolute maximum must occur at $t = 6$ or at an endpoint.

$$\begin{aligned} v(6) &= 55 + \int_0^6 a(t) dt \\ &= 55 + 2(15) + \frac{1}{2}(4)(15) = 115 > v(0) \\ \int_6^{18} a(t) dt &< 0 \text{ so } v(18) < v(6) \end{aligned}$$

Part D

1 point is earned for the answer

1 point is earned for the reason

The car's velocity is never equal to 0. The absolute minimum occurs at $t = 16$ where

$$v(6) = 115 + \int_6^{16} a(t) dt = 115 - 105 = 10 > 0.$$



0	1	2
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The student response earns all of the following points:

1 point is earned for the answer

1 point is earned for the reason

The car's velocity is never equal to 0. The absolute minimum occurs at $t = 16$ where

$$v(6) = 115 + \int_6^{16} a(t) dt = 115 - 105 = 10 > 0.$$

4-2-2a

22.

t (hours)	0	0.4	0.8	1.2	1.6	2.0	2.4
$v(t)$ (miles per hour)	0	11.8	9.5	17.2	16.3	16.8	20.1

Ruth rode her bicycle on a straight trail. She recorded her velocity $v(t)$, in miles per hour, for selected values of t over the interval $0 \leq t \leq 2.4$, as shown in the table above. For $0 < t \leq 2.4$, $v(t) > 0$.



According to the model, $g(t) = \frac{24t + 5 \sin(6t)}{t + 0.7}$, is Ruth's speed increasing or decreasing at time $t = 1.3$? Give a reason for your answer.

Part D

The response can earn up to 2 points:

2 points: For correct conclusion with reason

$$\text{Velocity} = g(1.3) = 18.096358 > 0$$

$$\text{Acceleration} = g'(1.3) = 3.761152 > 0$$

Ruth's speed is increasing at time $t = 1.3$ since velocity and acceleration have the same sign at this time.



0	1	2
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The response earns both of the following points:

2 points: For correct conclusion with reason

$$\text{Velocity} = g(1.3) = 18.096358 > 0$$

$$\text{Acceleration} = g'(1.3) = 3.761152 > 0$$

Ruth's speed is increasing at time $t = 1.3$ since velocity and acceleration have the same sign at this time.

4-2-2a

t (seconds)	0	10	40	60
$B(t)$ (meters)	100	136	9	49
$v(t)$ (meters per second)	2.0	2.3	2.5	4.6

Ben rides a unicycle back and forth along a straight east-west track. The twice-differentiable function B models Ben's position on the track, measured in meters from the western end of the track, at time t , measured in seconds from the start of the ride. The table above gives values for $B(t)$ and Ben's velocity, $v(t)$, measured in meters per second, at selected times t .

23. A light is directly above the western end of the track. Ben rides so that at time t , the distance $L(t)$ between Ben and the light satisfies $(L(t))^2 = 12^2 + (B(t))^2$. At what rate is the distance between Ben and the light changing at time $t=40$?

Part D

1 point is earned for the derivatives

1 point is earned if uses $B'(t) = v(t)$

1 point is earned for the answer

1 point is earned for units in (a) or (b)

$$2L(t)L'(t) = 2B(t)B'(t)$$

$$L'(40) = \frac{B(40)v(40)}{L(40)} = \frac{9 \cdot 2.5}{\sqrt{144+81}} = \frac{3}{2} \text{ meters/sec}$$



0	1	2	3	4
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The student response earns all of the following points:

1 point is earned for the derivatives

1 point is earned if uses $B'(t) = v(t)$

1 point is earned for the answer

1 point is earned for units in (a) or (b)

$$2L(t)L'(t) = 2B(t)B'(t)$$

4-2-2a

$$L'(40) = \frac{B(40)v(40)}{L(40)} = \frac{9 \cdot 2.5}{\sqrt{144+81}} = \frac{3}{2} \text{ meters/sec}$$

24. Use the data in the table to approximate Ben's acceleration at time $t=5$ seconds. Indicate units of measure.

Part A

1 point is earned for answer

$$a(5) \approx \frac{v(10)-v(0)}{10-0} = \frac{0.3}{10} = 0.03 \text{ meters/sec}^2$$



0	1
---	---

The student response earns one of the following points:

1 point is earned for answer

$$a(5) \approx \frac{v(10)-v(0)}{10-0} = \frac{0.3}{10} = 0.03 \text{ meters/sec}^2$$

4-2-2a

25.

t (seconds)	0	10	40	60
$B(t)$ (meters)	100	136	9	49
$v(t)$ (meters per second)	2.0	2.3	2.5	4.6

Ben rides a unicycle back and forth along a straight east-west track. The twice-differentiable function B models Ben's position on the track, measured in meters from the western end of the track, at time t , measured in seconds from the start of the ride. The table above gives values for $B(t)$ and Ben's velocity, $v(t)$, measured in meters per second, at selected times t .

(a) Use the data in the table to approximate Ben's acceleration at time $t = 5$ seconds. Indicate units of measure.

(b) Using correct units, interpret the meaning of $\int_0^{60} |v(t)| dt$ in the context of this problem.

Approximate $\int_0^{60} |v(t)| dt$ using a left Riemann sum with the subintervals indicated by the data in the table.

(c) For $40 \leq t \leq 60$, must there be a time t when Ben's velocity is 2 meters per second? Justify your answer.

(d) A light is directly above the western end of the track. Ben rides so that at time t , the distance $L(t)$ between Ben and the light satisfies $(L(t))^2 = 12^2 + (B(t))^2$. At what rate is the distance between Ben and the light changing at time $t = 40$?

Part A

1 point is earned for answer

$$a(5) \approx \frac{v(10) - v(0)}{10 - 0} = \frac{0.3}{10} = 0.03 \text{ meters/sec}^2$$

1 point is earned for units in (a) or (b)



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for answer

4-2-2a

$$a(5) \approx \frac{v(10) - v(0)}{10 - 0} = \frac{0.3}{10} = 0.03 \text{ meters/sec}^2$$

1 point is earned for units in (a) or (b)

Part B

1 point is earned for meaning of the integral

1 point is earned for the approximation

$\int_0^{60} |v(t)| dt$ is the total distance, in meters, that Ben rides over the 60-second interval $t = 0$ to $t = 60$.

$$\int_0^{60} |v(t)| dt \approx 2.0 \cdot 10 + 2.3(40 - 10) + 2.5(60 - 40) = 139 \text{ meters}$$

1 point is earned for units in (a) or (b)



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for meaning of the integral

1 point is earned for the approximation

$\int_0^{60} |v(t)| dt$ is the total distance, in meters, that Ben rides over the 60-second interval $t = 0$ to $t = 60$.

$$\int_0^{60} |v(t)| dt \approx 2.0 \cdot 10 + 2.3(40 - 10) + 2.5(60 - 40) = 139 \text{ meters}$$

1 point is earned for units in (a) or (b)

Part C

1 point is earned for difference quotient

4-2-2a

1 point is earned for conclusion with justification

Because $\frac{B(60) - B(40)}{60 - 40} = \frac{49 - 9}{20} = 2$, the Mean Value Theorem implies there is a time t , $40 < t < 60$, such that $v(t) = 2$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for difference quotient

1 point is earned for conclusion with justification

Because $\frac{B(60) - B(40)}{60 - 40} = \frac{49 - 9}{20} = 2$, the Mean Value Theorem implies there is a time t , $40 < t < 60$, such that $v(t) = 2$.

Part D

1 point is earned for the derivatives

1 point is earned if uses $B'(t) = v(t)$

1 point is earned for the answer

$$2L(t)L'(t) = 2B(t)B'(t)$$

$$L'(40) = \frac{B(40)v(40)}{L(40)} = \frac{9 \cdot 2.5}{\sqrt{144+81}} = \frac{3}{2} \text{ meters/sec}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the derivatives

1 point is earned if uses $B'(t) = v(t)$

1 point is earned for the answer

4-2-2a

$$2L(t)L'(t) = 2B(t)B'(t)$$

$$L'(40) = \frac{B(40)v(40)}{L(40)} = \frac{9 \cdot 2.5}{\sqrt{144+81}} = \frac{3}{2} \text{ meters/sec}$$

26. If the position of a particle on the x -axis at time t is $-5t^2$, then the average velocity of the particle for $0 \leq t \leq 3$ is

(A) -45

(B) -30

(C) -15



(D) -10

(E) -5

27. A particle moves along the x -axis so that at time $t \geq 0$ its position is given by $x(t) = 2t^3 - 21t^2 + 72t - 53$. At what time t is the particle at rest?

(A) $t = 1$ only

(B) $t = 3$ only

(C) $t = 7/2$ only

(D) $t = 3$ and $t = 7/2$

(E) $t = 3$ and $t = 4$



4-2-2a

28. A particle moves along the x -axis so that its velocity at time t , $0 \leq t \leq 5$, is given by $v(t) = 3(t-1)(t-3)$. At time $t=2$, the position of the particle is $x(2)=0$.

Find the minimum acceleration of the particle

Part A

1 point is earned for finding $a(t)$

1 point is earned for recognizing min at $t=0$

1 point is earned for evaluation

$$v(t) = 3t^2 - 12t + 9$$

$$a(t) = 6t - 12$$

a is increasing, so a is minimum at $t = 0$

$$a(0) = -12 \text{ is minimum value of } a.$$



0	1	2	3
---	---	---	---

The student response earns three of the following points:

1 point is earned for finding $a(t)$

1 point is earned for recognizing min at $t=0$

1 point is earned for evaluation

$$v(t) = 3t^2 - 12t + 9$$

$$a(t) = 6t - 12$$

a is increasing, so a is minimum at $t = 0$

$$a(0) = -12 \text{ is minimum value of } a.$$

4-2-2a

29. A particle moves along the x -axis so that its velocity at time t , $0 \leq t \leq 5$, is given by $v(t) = 3(t - 1)(t - 3)$. At time $t = 2$, the position of the particle is $x(2) = 0$.
- Find the minimum acceleration of the particle.
 - Find the total distance traveled by the particle.
 - Find the average velocity of the particle over the interval $0 \leq t \leq 5$.

Part A

1 point is earned for finding $a(t)$

1 point is earned for recognizing min at $t = 0$

1 point is earned for evaluation

$$v(t) = 3t^2 - 12t + 9$$

$$a(t) = 6t - 12$$

a is increasing, so a is minimum at $t = 0$

$a(0) = -12$ is minimum value of a .



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for finding $a(t)$

1 point is earned for recognizing min at $t = 0$

1 point is earned for evaluation

$$v(t) = 3t^2 - 12t + 9$$

$$a(t) = 6t - 12$$

a is increasing, so a is minimum at $t = 0$

$a(0) = -12$ is minimum value of a .

Part B

2 points are earned for integral expression

4-2-2a

1 point is earned for anti differentiation

1 point is earned for evaluation

Method 1



$$\begin{aligned}
 d &= \int_0^1 (3t^2 - 12t + 9)dt - \int_1^3 (3t^2 - 12t + 9)dt + \int_3^5 (3t^2 - 12t + 9)dt \\
 &= [t^3 - 6t^2 + 9t]_0^1 - [t^3 - 6t^2 + 9t]_1^3 + [t^3 - 6t^2 + 9t]_3^5 \\
 &= 4 - (-4) + 20 = 28
 \end{aligned}$$

or

Method 2

$$x(t) = t^3 - 6t^2 + 9t - 2$$

$$[(\text{or}) x(t) = t^3 - 6t^2 + 9t + C]$$

$$x(0) = -2$$

$$x(1) = 2$$

$$x(3) = -2$$

$$x(5) = 18$$

$$\text{Total distance} = 4 + 4 + 20 = 28$$



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

2 points are earned for integral expression

1 point is earned for anti differentiation

1 point is earned for evaluation

4-2-2a

Method 1



$$\begin{aligned}
 d &= \int_0^1 (3t^2 - 12t + 9)dt - \int_1^3 (3t^2 - 12t + 9)dt + \int_3^5 (3t^2 - 12t + 9)dt \\
 &= [t^3 - 6t^2 + 9t]_0^1 - [t^3 - 6t^2 + 9t]_1^3 + [t^3 - 6t^2 + 9t]_3^5 \\
 &= 4 - (-4) + 20 = 28
 \end{aligned}$$

or

Method 2

$$x(t) = t^3 - 6t^2 + 9t - 2$$

$$[(\text{or}) x(t) = t^3 - 6t^2 + 9t + C]$$

$$x(0) = -2$$

$$x(1) = 2$$

$$x(3) = -2$$

$$x(5) = 18$$

$$\text{Total distance} = 4 + 4 + 20 = 28$$

Part C

1 point is earned for applying definition of average velocity

1 point is earned for evaluation

Method 1

$$\begin{aligned}
 &\frac{\int_0^5 (3t^2 - 12t + 9)dt}{5 - 0} \\
 &= \frac{1}{5} [t^3 - 6t^2 + 9t]_0^5 = \frac{1}{5}(20) = 4
 \end{aligned}$$

or

4-2-2a

Method 2

$$\frac{x(5)-x(0)}{5-0} = \frac{18-(-2)}{5} = 4$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for applying definition of average velocity

1 point is earned for evaluation

Method 1

$$\frac{\int_0^5 (3t^2 - 12t + 9) dt}{5-0}$$

$$= \frac{1}{5} [t^3 - 6t^2 + 9t]_0^5 = \frac{1}{5} (20) = 4$$

or

Method 2

$$\frac{x(5)-x(0)}{5-0} = \frac{18-(-2)}{5} = 4$$

4-2-2a

A particle moves along the y -axis so that its velocity at any time $t \geq 0$ is given by $v(t) = t \cos t$. At time $t = 0$, the position of the particle is $y = 3$.

30. Write an expression for the acceleration of the particle in terms of t .

Part B

1 point is earned for correctly indicating $a(t) = v'(t)$

1 point is earned for the correct derivative

$$a(t) = v'(t) = \cos t - t \sin t$$



0	1	2
---	---	---

The student response earns two of the following points:

1 point is earned for correctly indicating $a(t) = v'(t)$

1 point is earned for the correct derivative

$$a(t) = v'(t) = \cos t - t \sin t$$

4-2-2a

31. For what values of t , $0 \leq t \leq 5$, is the particle moving upward?

Part A

1 point is earned for correctly setting $v(t) > 0$

$$t \cos t > 0, (\text{ or } v(t) > 0)$$

1 point is earned for the first correct interval

1 point is earned for the second correct interval

Particle is moving up for $0 < t < \frac{\pi}{2}$ and for $\frac{3\pi}{2} < t \leq 5$.

0/3 if uses any function other than $v(t)$



0	1	2	3
---	---	---	---

The student response earns two of the following points:

1 point is earned for correctly setting $v(t) > 0$

$$t \cos t > 0, (\text{ or } v(t) > 0)$$

1 point is earned for the first correct interval

1 point is earned for the second correct interval

Particle is moving up for $0 < t < \frac{\pi}{2}$ and for $\frac{3\pi}{2} < t \leq 5$

0/3 if uses any function other than $v(t)$

4-2-2a

32. A particle moves along the y -axis so that its velocity at any time $t \geq 0$ is given by $v(t) = t \cos t$. At time $t = 0$, the position of the particle is $y = 3$.
- For what values of t , $0 \leq t \leq 5$, is the particle moving upward?
 - Write an expression for the acceleration of the particle in terms of t .
 - Write an expression for the position $y(t)$ of the particle.
 - For $t > 0$, find the position of the particle the first time the velocity of the particle is zero.

Part A

1 point is earned for correctly setting $v(t) > 0$

$t \cos t > 0$, (or $v(t) > 0$)

1 point is earned for the first correct interval

1 point is earned for the second correct interval

Particle is moving up for $0 < t < \frac{\pi}{2}$ and for $\frac{3\pi}{2} < t \leq 5$.

0/3 if uses any function other than $v(t)$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for correctly setting $v(t) > 0$

$t \cos t > 0$, (or $v(t) > 0$)

1 point is earned for the first correct interval

1 point is earned for the second correct interval

Particle is moving up for $0 < t < \frac{\pi}{2}$ and for $\frac{3\pi}{2} < t \leq 5$.

0/3 if uses any function other than $v(t)$

Part B

1 point is earned for correctly indicating $a(t) = v'(t)$

4-2-2a

1 point is earned for the correct derivative

$$a(t) = v'(t) = \cos t - t \sin t$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for correctly indicating $a(t) = v'(t)$

1 point is earned for the correct derivative

$$a(t) = v'(t) = \cos t - t \sin t$$

Part C

1 point is earned for correctly indicating $y(t) = \int v(t)$

$$y(t) = \int v(t) dt$$

$$u = t \quad dv = \cos t \, dt$$

$$du = dt \quad v = \sin t$$

1 point is earned for the correct antiderivation (with or without C)

1 point is earned for the correct answer $C = 2$

$$y(t) = \int t \cos t \, dt = t \sin t - \int \sin t \, dt$$

$$= t \sin t + \cos t + C$$

$$3 = 1 + C; \, C = 2$$

$$y(t) = t \sin t + \cos t + 2$$

To be eligible for the third point, antiderivative must have at least 2 terms and at least 1 trig term



0	1	2	3
---	---	---	---

The student response earns all of the following points:

4-2-2a

1 point is earned for correctly indicating $y(t) = \int v(t)$

$$y(t) = \int v(t) dt$$

$$u = t \quad dv = \cos t \, dt$$

$$du = dt \quad v = \sin t$$

1 point is earned for the correct antiderivation (with or without C)

1 point is earned for the correct answer $C = 2$

$$y(t) = \int t \cos t \, dt = t \sin t - \int \sin t \, dt$$

$$= t \sin t + \cos t + C$$

$$3 = 1 + C; \, C = 2$$

$$y(t) = t \sin t + \cos t + 2$$

To be eligible for the third point, antiderivative must have at least 2 terms and at least 1 trig term

Part D

1 point is earned for the correct answer

Any non-constant function from part (c) is eligible for point.

$$y\left(\frac{\pi}{2}\right) = \frac{\pi}{2} + 2$$



0	1
---	---

The student response earns all of the following points:

1 point is earned for the correct answer

Any non-constant function from part (c) is eligible for point.

$$y\left(\frac{\pi}{2}\right) = \frac{\pi}{2} + 2$$

4-2-2a

33. Particle X moves along the positive x -axis so that its position at time $t \geq 0$ is given by $x(t) = 5t^3 - 9t^2 + 7$.
- (a) Is particle X moving toward the left or toward the right at time $t = 1$? Give a reason for your answer.
- (b) At what time $t \geq 0$ is particle X farthest to the left? Justify your answer.
- (c) A second particle, Y , moves along the positive y -axis so that its position at time t is given by $y(t) = 7t + 3$. At any time t , $t \geq 0$, the origin and the positions of the particles X and Y are the vertices of a triangle in the first quadrant. Find the rate of change of the area of the triangle at time $t = 1$. Show the work that leads to your answer.

Part A

The response can earn up to 2 points:

1 point: For consideration of $x'(1)$

1 point: For correct answer with reason

$$x'(t) = 15t^2 - 18t$$

$$x'(1) = 15 - 18 = -3$$

Since $x'(1) < 0$, the particle is moving to the left at time $t = 1$.



0	1	2
---	---	---

The response earns all of the following points:

1 point: For consideration of $x'(1)$

1 point: For correct answer with reason

$$x'(t) = 15t^2 - 18t$$

$$x'(1) = 15 - 18 = -3$$

Since $x'(1) < 0$, the particle is moving to the left at time $t = 1$.

Part B

The response can earn up to 3 points:

1 point: For consideration of $x'(t) = 0$

1 point: For identification of $t = \frac{6}{5}$

4-2-2a

1 point: For correct answer with reason

$$x'(t) = 3t(5t - 6) = 0 \Rightarrow t = 0, t = \frac{6}{5}$$

Since $x'(t) < 0$, for $0 < t < \frac{6}{5}$ and $x'(t) > 0$ for $t > \frac{6}{5}$, the particle is farthest to the left at time $t = \frac{6}{5}$.



0	1	2	3
---	---	---	---

The response earns all of the following points:

1 point: For consideration of $x'(t) = 0$

1 point: For identification of $t = \frac{6}{5}$

1 point: For correct answer with reason

$$x'(t) = 3t(5t - 6) = 0 \Rightarrow t = 0, t = \frac{6}{5}$$

Since $x'(t) < 0$, for $0 < t < \frac{6}{5}$ and $x'(t) > 0$ for $t > \frac{6}{5}$, the particle is farthest to the left at time $t = \frac{6}{5}$.

Part C

The response can earn up to 4 points:

1 point: For the area function

$$\begin{aligned} \text{Area} &= A(t) = \frac{1}{2}x(t)y(t) \\ &= \frac{1}{2}(5t^3 - 9t^2 + 7)(7t + 3) \end{aligned}$$

2 points: For the derivative

$$A'(t) = \frac{1}{2}[(15t^2 - 18t)(7t + 3) + (5t^3 - 9t^2 + 6)(7)]$$

1 point: For the correct answer

$$A'(1) = \frac{1}{2}[(-3)(10) + (3)(7)] = \frac{1}{2}[-30 + 21] = -\frac{9}{2}$$



0	1	2	3	4
---	---	---	---	---

The response earns all of the following points:

4-2-2a

1 point: For the area function

$$\begin{aligned}\text{Area} &= A(t) = \frac{1}{2}x(t)y(t) \\ &= \frac{1}{2}(5t^3 - 9t^2 + 7)(7t + 3)\end{aligned}$$

2 points: For the derivative

$$A'(t) = \frac{1}{2}[(15t^2 - 18t)(7t + 3) + (5t^3 - 9t^2 + 6)(7)]$$

1 point: For the correct answer

$$A'(1) = \frac{1}{2}[(-3)(10) + (3)(7)] = \frac{1}{2}[-30 + 21] = -\frac{9}{2}$$

4-2-2a

A particle moves along the x -axis so that its velocity v at time t , for $0 \leq t \leq 5$, is given by $v(t) = \ln(t^2 - 3t + 3)$. The particle is at position $x=8$ at time $t=0$.

34.  Find all times t in the open interval $0 < t < 5$ at which the particle changes direction. During which time intervals, for $0 \leq t \leq 5$, does the particle travel to the left?

Part B

1 point is earned for the sets $v(t)=0$

1 point is earned for the direction change at $t=1, 2$

1 point is earned for the interval with reason

$$v(t)=0$$

$$t^2 - 3t + 3 = 1$$

$$t^2 - 3t + 2 = 0$$

$$(t-2)(t-1)=0$$

$$t=1, 2$$

$$v(t)>0 \text{ for } 0 < t < 1$$

$$v(t)<0 \text{ for } 1 < t < 2$$

$$v(t)>0 \text{ for } 2 < t < 5$$

The particle changes direction when $t=1$ and $t=2$.

The particle travels to the left when $1 < t < 2$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the sets $v(t)=0$

1 point is earned for the direction change at $t=1, 2$

1 point is earned for the interval with reason

$$v(t)=0$$

4-2-2a

$$t^2 - 3t + 3 = 1$$

$$t^2 - 3t + 2 = 0$$

$$(t-2)(t-1)=0$$

$$t=1, 2$$

$$v(t)>0 \text{ for } 0<t<1$$

$$v(t)<0 \text{ for } 1<t<2$$

$$v(t)>0 \text{ for } 2<t<5$$

The particle changes direction when $t=1$ and $t=2$.

The particle travels to the left when $1<t<2$.

35.  (a) Find the acceleration of the particle at time $t = 4$.

Part A

1 point is earned for the answer

$$a(4)=v'(4)=\frac{5}{7}$$



0	1
---	---

The student response earns one of the following points:

1 point is earned for the answer

$$a(4)=v'(4)=\frac{5}{7}$$

4-2-2a

36.  A particle moves along the x -axis so that its velocity v at time t , for $0 \leq t \leq 5$, is given by $v(t) = \ln(t^2 - 3t + 3)$. The particle is at position $x = 8$ at time $t = 0$.

- (a) Find the acceleration of the particle at time $t = 4$.
- (b) Find all times t in the open interval $0 < t < 5$ at which the particle changes direction. During which time intervals, for $0 \leq t \leq 5$, does the particle travel to the left?
- (c) Find the position of the particle at time $t = 2$.
- (d) Find the average speed of the particle over the interval $0 \leq t \leq 2$.

Part A

1 point is earned for the answer

$$a(4) = v'(4) = \frac{5}{7}$$



0	1
---	---

The student response earns all of the following points:

1 point is earned for the answer

$$a(4) = v'(4) = \frac{5}{7}$$

Part B

1 point is earned for setting $v(t) = 0$

1 point is earned for the direction change at $t = 1, 2$

1 point is earned for the interval with reason

$$v(t) = 0$$

$$t^2 - 3t + 3 = 1$$

$$t^2 - 3t + 2 = 0$$

$$(t - 2)(t - 1) = 0$$

$$t = 1, 2$$

$$v(t) > 0 \text{ for } 0 < t < 1$$

$$v(t) < 0 \text{ for } 1 < t < 2$$

4-2-2a

$$v(t) > 0 \text{ for } 2 < t < 5$$

The particle changes direction when $t = 1$ and $t = 2$.

The particle travels to the left when $1 < t < 2$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for setting $v(t) = 0$

1 point is earned for the direction change at $t = 1, 2$

1 point is earned for the interval with reason

$$v(t) = 0$$

$$t^2 - 3t + 3 = 1$$

$$t^2 - 3t + 2 = 0$$

$$(t - 2)(t - 1) = 0$$

$$t = 1, 2$$

$$v(t) > 0 \text{ for } 0 < t < 1$$

$$v(t) < 0 \text{ for } 1 < t < 2$$

$$v(t) > 0 \text{ for } 2 < t < 5$$

The particle changes direction when $t = 1$ and $t = 2$.

The particle travels to the left when $1 < t < 2$.

Part C

1 point is earned for $\int_0^2 \ln(u^2 - 3u + 3) du$

1 point is earned if handles initial condition

1 point is earned for answer

4-2-2a

$$s(t) = s(0) + \int_0^t \ln(u^2 - 3u + 3) du$$

$$s(2) = 8 + \int_0^2 \ln(u^2 - 3u + 3) du$$

$$= 8.368 \text{ or } 8.369$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for $\int_0^2 \ln(u^2 - 3u + 3) du$

1 point is earned if handles initial condition

1 point is earned for answer

$$s(t) = s(0) + \int_0^t \ln(u^2 - 3u + 3) du$$

$$s(2) = 8 + \int_0^2 \ln(u^2 - 3u + 3) du$$

$$= 8.368 \text{ or } 8.369$$

Part D

1 point is earned for the integral

1 point is earned for the answer

$$\frac{1}{2} \int_0^2 |v(t)| dt = 0.370 \text{ or } 0.371$$



0	1	2
---	---	---

4-2-2a

The student response earns all of the following points:

1 point is earned for the integral

1 point is earned for the answer

$$\frac{1}{2} \int_0^2 |v(t)| dt = 0.370 \text{ or } 0.371$$

4-2-2a

37. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (hours)	0	1	2	3	4
$B(t)$ (miles per hour)	1	8	1.5	-5	11

Brandon and Chloe ride their bikes for 4 hours along a flat, straight road. Brandon's velocity, in miles per hour, at time t hours is given by a differentiable function B for $0 \leq t \leq 4$. Values of $B(t)$ for selected times t are given in the table above. Chloe's velocity, in miles per hour, at time t hours is given by the piecewise function C defined by

$$C(t) = \begin{cases} te^{4-t^2} & \text{for } 0 \leq t \leq 2 \\ 12 - 3t - t^2 & \text{for } 2 < t \leq 4. \end{cases}$$

- How many miles did Chloe travel from time $t = 0$ to time $t = 2$?
- At time $t = 3$, is Chloe's speed increasing or decreasing? Give a reason for your answer.
- Is there a time t , for $0 \leq t \leq 4$, at which Brandon's acceleration is equal to 2.5 miles per hour per hour? Justify your answer.
- Is there a time t , for $0 \leq t \leq 2$, at which Brandon's velocity is equal to Chloe's velocity? Justify your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned with a connection to the correct part of the piecewise defined function. $\int_0^2 C(t) dt$ by itself is not sufficient to earn the point.

The second point is earned with $-\frac{1}{2}e^{4-t^2}$ or with $-\frac{1}{2}e^u$ for $u = 4 - t^2$. Limits of integration are part of the third point.

The third point does not require units.

4-2-2a



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ integral
- ☐ antiderivative
- ☐ answer

Solution:

$$\int_0^2 te^{4-t^2} dt = -\frac{1}{2}e^{4-t^2} \Big|_{t=0}^{t=2} = -\frac{1}{2} + \frac{1}{2}e^4$$

Chloe traveled $-\frac{1}{2} + \frac{1}{2}e^4$ miles from time $t = 0$ to time $t = 2$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned by showing both $C(3)$ and $C'(3)$ are less than zero or that both have the same sign.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ $C(3) < 0$ and $C'(3) < 0$
- ☐ answer with reason

Solution:

$$C(3) = -6 < 0$$

$$C'(3) = -9 < 0$$

Chloe's speed is increasing at time $t = 3$ because her velocity and acceleration have the same sign.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

4-2-2a

The second point is earned with the justification that B is continuous and a numerical difference quotient equal to 2.5. There is no requirement for the response to indicate that B is differentiable, nor to refer to the Mean Value Theorem by name.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ $\frac{B(4)-B(0)}{4-0}$
- ☐ answer with justification

Solution:

B is differentiable $\Rightarrow B$ is continuous on $[0, 4]$.

$$\frac{B(4)-B(0)}{4-0} = \frac{11-1}{4-0} = 2.5$$

By the Mean Value Theorem, there is a time t , for $0 < t < 4$, such that $B'(t) = 2.5$ miles per hour per hour.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned by comparing B and C , either by a subtraction statement or by comparing $B(t)$ and $C(t)$ for some t in $\{0, 1, 2\}$.

The second point is earned with a justification that includes asserting the continuity of B and C or of $B - C$, and showing that $B(0) > C(0)$ and $B(2) < C(2)$. It is not required for the response to specifically refer to the Intermediate Value Theorem.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ considers $B(t) - C(t)$
- ☐ answer with justification

Solution:

B and C are continuous on $[0, 2]$, therefore $B - C$ is continuous on $[0, 2]$.

4-2-2a

$$B(0) - C(0) = 1 - 0 > 0$$

$$B(2) - C(2) = 1.5 - 2 < 0$$

By the Intermediate Value Theorem, there is a time t , for $0 < t < 2$, such that $B(t) - C(t) = 0$, or $B(t) = C(t)$.

Two particles move along the x -axis. For $0 \leq t \leq 6$, the position of particle P at time t is given by $p(t) = 2\cos(\pi/4t)$, while the position of particle R at time t is given by $r(t) = t^3 - 6t^2 + 9t + 3$.

38. Find the acceleration of particle P at time $t=3$. Is particle P speeding up, slowing down, or doing neither at time $t=3$? Explain your reasoning.

Part C

1 point is earned for $p''(3)$

1 point is earned for answer with reason

$$p''(t) = -2 \cdot \frac{\pi}{4} \cdot \frac{\pi}{4} \cos\left(\frac{\pi}{4}t\right)$$

$$p''(3) = -2\left(\frac{\pi}{4}\right)^2 \cos\left(\frac{3\pi}{4}\right) = \frac{\pi^2}{8} \cdot \frac{\sqrt{2}}{2} > 0$$

$$p'(3) < 0$$

Therefore particle P is slowing down at time $t=3$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $p''(3)$

1 point is earned for answer with reason

$$p''(t) = -2 \cdot \frac{\pi}{4} \cdot \frac{\pi}{4} \cos\left(\frac{\pi}{4}t\right)$$

$$p''(3) = -2\left(\frac{\pi}{4}\right)^2 \cos\left(\frac{3\pi}{4}\right) = \frac{\pi^2}{8} \cdot \frac{\sqrt{2}}{2} > 0$$

$$p'(3) < 0$$

Therefore particle P is slowing down at time $t=3$.

4-2-2a

39. For $0 \leq t \leq 6$, find all times t during which the two particles travel in opposite directions.

Part B

1 point is earned for $p'(t)$

1 point is earned for sign analysis for $p'(t)$

1 point is earned for the answer

$$p'(t) = -2 \cdot \frac{\pi}{4} \sin\left(\frac{\pi}{4}t\right)$$

$$p'(T) = 0 \text{ when } t = 0 \text{ and } t = 4$$

$$p'(t) < 0 \text{ for } 0 < t < 4$$

$$p'(t) > 0 \text{ for } 4 < t < 6$$

Therefore the particles travel in opposite directions for $0 < t < 1$ and $3 < t < 4$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for $p'(t)$

1 point is earned for sign analysis for $p'(t)$

1 point is earned for the answer

$$p'(t) = -2 \cdot \frac{\pi}{4} \sin\left(\frac{\pi}{4}t\right)$$

$$p'(T) = 0 \text{ when } t = 0 \text{ and } t = 4$$

$$p'(t) < 0 \text{ for } 0 < t < 4$$

$$p'(t) > 0 \text{ for } 4 < t < 6$$

Therefore the particles travel in opposite directions for $0 < t < 1$ and $3 < t < 4$.

4-2-2a

40. For $0 \leq t \leq 6$, find all times t during which particle R is moving to the right.
-

Part A

1 point is earned for $r'(t)$

1 point is earned for the answer

$$r'(t) = 3t^2 - 12t + 9 = 3(t - 1)(t - 3)$$

$$r'(t) = 0 \text{ when } t = 1 \text{ and } t = 3$$

$$r'(t) > 0 \text{ for } 0 < t < 1 \text{ and } 3 < t < 6$$

$$r'(t) < 0 \text{ for } 1 < t < 3$$

$r'(t) < 0$ for $1 < t < 3$ Therefore R is moving to the right for $0 < t < 1$ and $3 < t < 6$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $r'(t)$

1 point is earned for the answer

$$r'(t) = 3t^2 - 12t + 9 = 3(t - 1)(t - 3)$$

$$r'(t) = 0 \text{ when } t = 1 \text{ and } t = 3$$

$$r'(t) > 0 \text{ for } 0 < t < 1 \text{ and } 3 < t < 6$$

$$r'(t) < 0 \text{ for } 1 < t < 3$$

Therefore R is moving to the right for $0 < t < 1$ and $3 < t < 6$.

4-2-2a

41. Two particles move along the x -axis. For $0 \leq t \leq 6$, the position of particle P at time t is given by $p(t) = 2\cos(\pi/4t)$, while the position of particle R at time t is given by $r(t) = t^3 - 6t^2 + 9t + 3$.
- (a) For $0 \leq t \leq 6$, find all times t during which particle R is moving to the right.
- (b) For $0 \leq t \leq 6$, find all times t during which the two particles travel in opposite directions.
- (c) Find the acceleration of particle P at time $t = 3$. Is particle P speeding up, slowing down, or doing neither at time $t = 3$? Explain your reasoning.
- (d) Write, but do not evaluate, an expression for the average distance between the two particles on the interval $1 \leq t \leq 3$.

Part A

1 point is earned for $r'(t)$

1 point is earned for the answer

$$r'(t) = 3t^2 - 12t + 9 = 3(t - 1)(t - 3)$$

$$r'(t) = 0 \text{ when } t = 1 \text{ and } t = 3$$

$$r'(t) > 0 \text{ for } 0 < t < 1 \text{ and } 3 < t < 6$$

$$r'(t) < 0 \text{ for } 1 < t < 3$$

Therefore R is moving to the right for $0 < t < 1$ and $3 < t < 6$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $r'(t)$

1 point is earned for the answer

$$r'(t) = 3t^2 - 12t + 9 = 3(t - 1)(t - 3)$$

$$r'(t) = 0 \text{ when } t = 1 \text{ and } t = 3$$

$$r'(t) > 0 \text{ for } 0 < t < 1 \text{ and } 3 < t < 6$$

$$r'(t) < 0 \text{ for } 1 < t < 3$$

Therefore R is moving to the right for $0 < t < 1$ and $3 < t < 6$.

4-2-2a

Part B

1 point is earned for $p'(t)$

1 point is earned for sign analysis for $p'(t)$

1 point is earned for the answer

$$p'(t) = -2 \cdot \frac{\pi}{4} \sin\left(\frac{\pi}{4}t\right)$$

$$p'(t) = 0 \text{ when } t = 0 \text{ and } t = 4$$

$$p'(t) < 0 \text{ for } 0 < t < 4$$

$$p'(t) > 0 \text{ for } 4 < t < 6$$

Therefore the particles travel in opposite directions for $0 < t < 1$ and $3 < t < 4$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for $p'(t)$

1 point is earned for sign analysis for $p'(t)$

1 point is earned for the answer

$$p'(t) = -2 \cdot \frac{\pi}{4} \sin\left(\frac{\pi}{4}t\right)$$

$$p'(t) = 0 \text{ when } t = 0 \text{ and } t = 4$$

$$p'(t) < 0 \text{ for } 0 < t < 4$$

$$p'(t) > 0 \text{ for } 4 < t < 6$$

Therefore the particles travel in opposite directions for $0 < t < 1$ and $3 < t < 4$.

Part C

1 point is earned for $p''(3)$

4-2-2a

1 point is earned for answer with reason

$$p''(t) = -2 \cdot \frac{\pi}{4} \cdot \frac{\pi}{4} \cos\left(\frac{\pi}{4}t\right)$$

$$p''(3) = -2\left(\frac{\pi}{4}\right)^2 \cos\left(\frac{3\pi}{4}\right) = \frac{\pi^2}{8} \cdot \frac{\sqrt{2}}{2} > 0$$

$$p'(3) < 0$$

Therefore particle P is slowing down at time $t = 3$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $p''(3)$

1 point is earned for answer with reason

$$p''(t) = -2 \cdot \frac{\pi}{4} \cdot \frac{\pi}{4} \cos\left(\frac{\pi}{4}t\right)$$

$$p''(3) = -2\left(\frac{\pi}{4}\right)^2 \cos\left(\frac{3\pi}{4}\right) = \frac{\pi^2}{8} \cdot \frac{\sqrt{2}}{2} > 0$$

$$p'(3) < 0$$

Therefore particle P is slowing down at time $t = 3$.

Part D

1 point is earned for the integrand

1 point is earned for the limits and constant

$$\frac{1}{2} \int_1^3 |p(t) - r(t)| dt$$



0	1	2
---	---	---

The student response earns all of the following points:

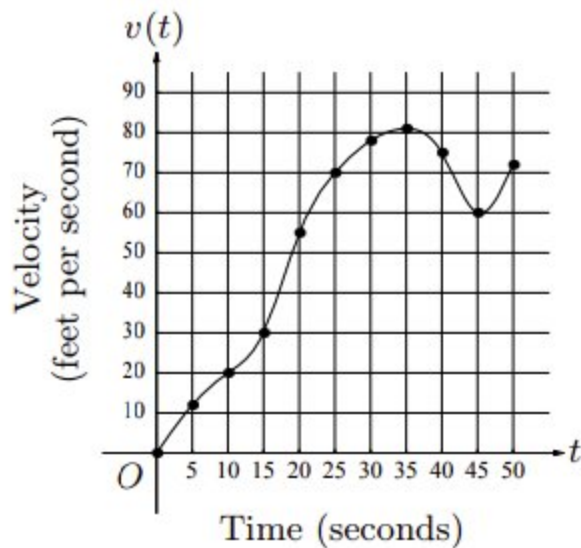
1 point is earned for the integrand

4-2-2a

1 point is earned for the limits and constant

$$\frac{1}{2} \int_1^3 |p(t) - r(t)| dt$$

4-2-2a



t (seconds)	$v(t)$ (feet per second)
0	0
5	12
10	20
15	30
20	55
25	70
30	78
35	81
40	75
45	60
50	72

The graph of the velocity $v(t)$, in ft/sec, of a car traveling on a straight road, for $0 \leq t \leq 50$, is shown above. A table of values for $v(t)$, at 5 second intervals of time t , is shown to the right of the graph

42. Find one approximation for the acceleration of the car, in ft/sec^2 , at $t = 40$. Show the computations you used to arrive at your answer.

Part C

1 point is earned for the method

1 point is earned for the answer

Difference quotient; e.g

$$\frac{v(45) - v(40)}{5} = \frac{60 - 75}{5} = -3 \text{ ft/sec}^2 \text{ or}$$

$$\frac{v(40) - v(35)}{5} = \frac{75 - 81}{5} = -\frac{6}{5} \text{ ft/sec}^2 \text{ or}$$

4-2-2a

$$\frac{v(40) - v(35)}{10} = \frac{60 - 81}{10} = -\frac{21}{10} \text{ ft/sec}^2$$



0	1	2
---	---	---

The student response earns two of the following points:

1 point is earned for the method

1 point is earned for the answer

Difference quotient; e.g

$$\frac{v(45) - v(40)}{5} = \frac{60 - 75}{5} = -3 \text{ ft/sec}^2 \text{ or}$$

$$\frac{v(40) - v(35)}{5} = \frac{75 - 81}{5} = -\frac{6}{5} \text{ ft/sec}^2 \text{ or}$$

$$\frac{v(40) - v(35)}{10} = \frac{60 - 81}{10} = -\frac{21}{10} \text{ ft/sec}^2$$

43. Find the average acceleration of the car, in ft/sec^2 , over the interval $0 \leq t \leq 50$.

Part B

1 point is earned for the answer

$$\text{Avg. Acc.} = \frac{v(50) - v(0)}{50 - 0} = \frac{72 - 0}{50} = \frac{72}{50}$$

or 1.44 ft/sec^2



0	1
---	---

The student response earns one of the following points:

1 point is earned for the answer

$$\text{Avg. Acc.} = \frac{v(50) - v(0)}{50 - 0} = \frac{72 - 0}{50} = \frac{72}{50}$$

or 1.44 ft/sec^2

4-2-2a

44. During what intervals of time is the acceleration of the car positive? Give a reason for your answer

Part A

1 point is earned for (0, 35)

1 point is earned for (45, 50)

1 point is earned for the reason

Acceleration is positive on (0, 35) and (45, 50) because the velocity $v(t)$ is increasing on $[0, 35]$ and $[45, 50]$



0	1	2	3
---	---	---	---

The student response earns three of the following points:

1 point is earned for (0, 35)

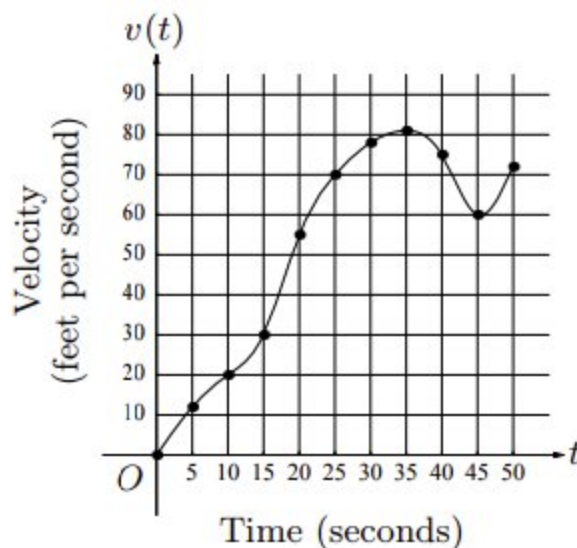
1 point is earned for (45, 50)

1 point is earned for the reason

Acceleration is positive on (0, 35) and (45, 50) because the velocity $v(t)$ is increasing on $[0, 35]$ and $[45, 50]$

4-2-2a

45.



t (seconds)	$v(t)$ (feet per second)
0	0
5	12
10	20
15	30
20	55
25	70
30	78
35	81
40	75
45	60
50	72

The graph of the velocity $v(t)$, in ft/sec, of a car traveling on a straight road, for $0 \leq t \leq 50$, is shown above. A table of values for $v(t)$, at 5 second intervals of time t , is shown to the right of the graph.

(a) During what intervals of time is the acceleration of the car positive? Give a reason for your answer.

(b) Find the average acceleration of the car, in ft/sec^2 , over the interval $0 \leq t \leq 50$.

(c) Find one approximation for the acceleration of the car, in ft/sec^2 , at $t = 40$. Show the computations you used to arrive at your answer.

(d) Approximate $\int_0^{50} v(t) dt$ with a Riemann sum, using the midpoints of five subintervals of equal length. Using correct units, explain the meaning of this integral.

Part A

1 point is earned for (0, 35)

1 point is earned for (45, 50)

4-2-2a

1 point is earned for the reason

Acceleration is positive on $(0, 35)$ and $(45, 50)$ because the velocity $v(t)$ is increasing on $[0, 35]$ and $[45, 50]$



0	1	2	3
---	---	---	---

The student response earns three of the following points:

1 point is earned for $(0, 35)$

1 point is earned for $(45, 50)$

1 point is earned for the reason

Acceleration is positive on $(0, 35)$ and $(45, 50)$ because the velocity $v(t)$ is increasing on $[0, 35]$ and $[45, 50]$

Part B

1 point is earned for the answer

$$\text{Avg. Acc.} = \frac{v(50) - v(0)}{50 - 0} = \frac{72 - 0}{50} = \frac{72}{50}$$

or 1.44 ft/sec^2



0	1
---	---

The student response earns all of the following points:

1 point is earned for the answer

$$\text{Avg. Acc.} = \frac{v(50) - v(0)}{50 - 0} = \frac{72 - 0}{50} = \frac{72}{50}$$

or 1.44 ft/sec^2

Part C

4-2-2a

1 point is earned for the method

1 point is earned for the answer

Difference quotient; e.g

$$\frac{v(45) - v(40)}{5} = \frac{60 - 75}{5} = -3 \text{ ft/sec}^2 \text{ or}$$

$$\frac{v(40) - v(35)}{5} = \frac{75 - 81}{5} = -\frac{6}{5} \text{ ft/sec}^2 \text{ or}$$

$$\frac{v(40) - v(35)}{10} = \frac{60 - 81}{10} = -\frac{21}{10} \text{ ft/sec}^2$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the method

1 point is earned for the answer

Difference quotient; e.g

$$\frac{v(45) - v(40)}{5} = \frac{60 - 75}{5} = -3 \text{ ft/sec}^2 \text{ or}$$

$$\frac{v(40) - v(35)}{5} = \frac{75 - 81}{5} = -\frac{6}{5} \text{ ft/sec}^2 \text{ or}$$

$$\frac{v(40) - v(35)}{10} = \frac{60 - 81}{10} = -\frac{21}{10} \text{ ft/sec}^2$$

Part D

1 point is earned for midpoint Riemann sum

4-2-2a

1 point is earned for answer

1 point is earned for meaning of integral

$$\begin{aligned} \int_0^{50} v(t) dt \\ &\approx 10 [v(5) + v(15) + v(25) + v(35) + v(45)] \\ &= 10(12 + 30 + 70 + 81 + 60) \\ &= 2530 \text{ feet} \end{aligned}$$

This integral is the total distance traveled in feet over the time 0 to 50 seconds.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for midpoint Riemann sum

1 point is earned for answer

1 point is earned for meaning of integral

$$\begin{aligned} \int_0^{50} v(t) dt \\ &\approx 10 [v(5) + v(15) + v(25) + v(35) + v(45)] \\ &= 10(12 + 30 + 70 + 81 + 60) \\ &= 2530 \text{ feet} \end{aligned}$$

This integral is the total distance traveled in feet over the time 0 to 50 seconds.

4-2-2a

46. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Emma and Lily jog in the same direction along a straight track. For $0 \leq t \leq 15$, Emma's velocity at time t is given by $E(t) = \frac{7510}{t^2 - 7t + 80.22}$ and Lily's velocity at time t is given by $L(t) = 12t^3e^{-0.5t}$. Both $E(t)$ and $L(t)$ are positive for $0 \leq t \leq 15$ and are measured in meters per minute, and t is measured in minutes. Emma is 10 meters ahead of Lily at time $t = 0$, and Emma remains ahead of Lily for $0 < t \leq 15$.

- (a) Find the value of $\frac{1}{6} \int_2^8 E(t) dt$. Using correct units, interpret the meaning of $\frac{1}{6} \int_2^8 E(t) dt$ in the context of the problem.
- (b) At time $t = 5$, is Emma speeding up or slowing down? Give a reason for your answer.
- (c) Is the distance between Emma and Lily at time $t = 7$ increasing or decreasing? Give a reason for your answer.
- (d) What is the maximum distance between Emma and Lily over the time interval $0 \leq t \leq 15$? Justify your answer.

Part A

For the second point, the response must include reference to average value over the correct time interval with correct units.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

4-2-2a

The student response accurately includes both of the criteria below.

- ☐ value
- ☐ meaning

Solution:

$$\frac{1}{6} \int_2^8 E(t) dt = 103.178012$$

Emma's average velocity over the time interval $2 \leq t \leq 8$ is 103.178 meters per minute.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ uses $E(5)$ and $E'(5)$
- ☐ answer with reason

Solution:

$$E(5) > 0 \text{ and } E'(5) = -4.569193 < 0$$

At time $t = 5$, since Emma has positive velocity and negative acceleration, she is slowing down.

Part C

For the first point, $E(7) - L(7) < 0$ is required with or without numerical values.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1
---	---

The student response accurately includes a correct answer with reason.

Solution:

4-2-2a

Emma's velocity at $t = 7$ is $E(7) = 93.617552$.

Lily's velocity at time $t = 7$ is $L(7) = 124.292430$.

Since $E(7) - L(7) < 0$, the distance between Emma and Lily at $t = 7$ is decreasing.

Part D

The third point is earned with correct use of the Candidates Test.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3	4
---	---	---	---	---

The student response accurately includes all four of the criteria below.

- ☐ considers $E(t) - L(t) = 0$
- ☐ $t = 4.233$ and $t = 11.671$ as candidates
- ☐ justification
- ☐ answer

Solution:

The distance between Emma and Lily at time t is given by $10 + \int_0^t (E(x) - L(x)) dx$.

$$\frac{d}{dt} \left(10 + \int_0^t (E(x) - L(x)) dx \right) = E(t) - L(t) = 0$$

$$\Rightarrow t_1 = 4.232614 \text{ and } t_2 = 11.671403$$

t	Distance between Emma and Lily
0	10
t_1	266.998916
t_2	126.880323
15	156.143676

The maximum distance between Emma and Lily is 267 meters.

4-2-2a

47. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Mary and Chance walk in the same direction along a straight path. For $0 \leq t \leq 20$, Mary's velocity at time t is given by $M(t) = \frac{6010}{t^2 - 3t + 50.5}$ and Chance's velocity at time t is given by $C(t) = 8.5t^3 e^{-0.45t}$. Both $M(t)$ and $C(t)$ are positive for $0 \leq t \leq 20$ and are measured in meters per minute, and t is measured in minutes. Mary is 12 meters ahead of Chance at time $t = 0$, and Mary remains ahead of Chance for $0 < t \leq 20$.

- (a) Find the value of $\frac{1}{10} \int_5^{15} M(t) \, dt$. Using correct units, interpret the meaning of $\frac{1}{10} \int_5^{15} M(t) \, dt$ in the context of the problem.
- (b) At time $t = 10$, is Mary speeding up or slowing down? Give a reason for your answer.
- (c) Is the distance between Mary and Chance at time $t = 18$ increasing or decreasing? Give a reason for your answer.
- (d) What is the maximum distance between Mary and Chance over the time interval $0 \leq t \leq 20$? Justify your answer.

Part A

For the second point, the response must include reference to average value over the correct time interval with correct units.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

4-2-2a

The student response accurately includes both of the criteria below.

- ☐ value
- ☐ meaning

Solution:

$$\frac{1}{10} \int_5^{15} M(t) dt = 54.409167$$

Mary's average velocity over the time interval $5 \leq t \leq 10$ is 54.409 meters per minute.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ uses $M(10)$ and $M'(10)$
- ☐ answer with reason

Solution:

$$M(10) > 0 \text{ and } M'(10) = -7.036380 < 0$$

At time $t = 10$, since Mary has positive velocity and negative acceleration, she is slowing down.

Part C

For the point, $M(18) - C(18) > 0$ is required with or without numerical values.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1
---	---

The student response accurately includes a correct answer with reason.

Solution:

4-2-2a

Mary's velocity at time $t = 18$ is $M(18) = 18.751950$. Chance's velocity at time $t = 18$ is $C(18) = 15.047042$.

Since $M(18) - C(18) > 0$, the distance between Mary and Chance at time $t = 18$ is increasing.

Part D

The third point is earned with correct use of the Candidates Test.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3	4
---	---	---	---	---

The student response accurately includes all four of the criteria below.

- ☐ considers $M(t) - C(t) = 0$
- ☐ $t = 4.584$ and $t = 16.715$ as candidates
- ☐ justification
- ☐ answer

Solution:

The distance between Mary and Chance at time t is given by $12 + \int_0^t (M(x) - C(x)) dx$.

$$\frac{d}{dt} \left(12 + \int_0^t (M(x) - C(x)) dx \right) = M(t) - C(t) = 0$$

$$\Rightarrow t_1 = 4.583625 \text{ and } t_2 = 16.714620$$

t	Distance between Mary and Chance
0	12
t_1	365.527469
t_2	13.344315
20	27.010082

The maximum distance between Mary and Chance is 365.527 meters.

4-2-2a

48. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

A snail is traveling along a straight path. The snail's velocity can be modeled by $v(t) = 1.4 \ln(1 + t^2)$ inches per minute for $0 \leq t \leq 15$ minutes.



- (a) Find the acceleration of the snail at time $t = 5$ minutes.
- (b) What is the displacement of the snail over the interval $0 \leq t \leq 15$ minutes?
- (c) At what time t , $0 \leq t \leq 15$, is the snail's instantaneous velocity equal to its average velocity over the interval $0 \leq t \leq 15$?
- (d) An ant arrives at the snail's starting position at time $t = 12$ minutes and follows the snail's path. During the interval $12 \leq t \leq 15$ minutes, the ant travels in the same direction as the snail with a constant acceleration of 2 inches per minute per minute. The ant catches up to the snail at time $t = 15$ minutes. The ant's velocity at time $t = 12$ is B inches per minute. Find the value of B .

Part A

1 point is earned for: answer

$$v'(5) = 0.538462$$

The acceleration of the snail at time $t = 5$ minutes is 0.538 inches per minute per minute.



0	1
---	---

The student response earns all of the following points:

1 point is earned for: answer

$$v'(5) = 0.538462$$

The acceleration of the snail at time $t = 5$ minutes is 0.538 inches per minute per minute.

Part B

1 point is earned for: integral

1 point is earned for: answer

$$\int_0^{15} v(t) dt = 76.043074$$

The displacement of the snail over the interval $0 \leq t \leq 15$ minutes is 76.043 inches.

4-2-2a



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for: integral

1 point is earned for: answer

$$\int_0^{15} v(t) \, dt = 76.043074$$

The displacement of the snail over the interval $0 \leq t \leq 15$ minutes is 76.043 inches.

Part C

1 point is earned for: average value expression

1 point is earned for: answer

$$\frac{1}{15} \int_0^{15} v(t) \, dt = 5.069538$$

$$1.4 \ln(1 + t^2) = 5.069538 \Rightarrow t = 6.031 \text{ minutes}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for: average value expression

1 point is earned for: answer

$$\frac{1}{15} \int_0^{15} v(t) \, dt = 5.069538$$

$$1.4 \ln(1 + t^2) = 5.069538 \Rightarrow t = 6.031 \text{ minutes}$$

Part D

1 point is earned for: ant's velocity

1 point is earned for: ant's displacement

4-2-2a

1 point is earned for: equation**1 point is earned for:** answer

The velocity of the ant at time t , $12 \leq t \leq 15$, is $\int 2 \, dt = 2t + c$ inches per minute for some constant c .

For $12 \leq t \leq 15$, the displacement of the ant is $\int_{12}^{15} (2t + c) \, dt = (t^2 + ct) \bigg|_{t=12}^{t=15} = 81 + 3c$ inches.

Thus, $81 + 3c = 76.043074 \Rightarrow c = -1.652309$.

The velocity of the ant at time $t = 12$ is $B = 2 \cdot 12 - 1.652309 = 22.348$ (or 22.347) inches per minute.

— OR —

The velocity of the ant at time t , $12 \leq t \leq 15$, is $2(t - 12) + B$ inches per minute.

For $12 \leq t \leq 15$, the displacement of the ant is $\int_{12}^{15} (2(t - 12) + B) \, dt = ((t - 12)^2 + Bt) \bigg|_{t=12}^{t=15} = 9 + 3B$ inches.

$9 + 3B = 76.043074 \Rightarrow B = 22.348$ (or 22.347) inches per minute



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for: ant's velocity**1 point is earned for:** ant's displacement**1 point is earned for:** equation**1 point is earned for:** answer

The velocity of the ant at time t , $12 \leq t \leq 15$, is $\int 2 \, dt = 2t + c$ inches per minute for some constant c .

For $12 \leq t \leq 15$, the displacement of the ant is $\int_{12}^{15} (2t + c) \, dt = (t^2 + ct) \bigg|_{t=12}^{t=15} = 81 + 3c$ inches.

Thus, $81 + 3c = 76.043074 \Rightarrow c = -1.652309$.

4-2-2a

The velocity of the ant at time $t = 12$ is $B = 2 \cdot 12 - 1.652309 = 22.348$ (or 22.347) inches per minute.

— OR —

The velocity of the ant at time t , $12 \leq t \leq 15$, is $2(t - 12) + B$ inches per minute.

For $12 \leq t \leq 15$, the displacement of the ant is $\int_{12}^{15} (2(t - 12) + B) dt = \left((t - 12)^2 + Bt \right) \bigg|_{t=12}^{t=15} = 9 + 3B$ inches.

$$9 + 3B = 76.043074 \Rightarrow B = 22.348 \text{ (or 22.347) inches per minute}$$

49. A particle moves along the x-axis so that its acceleration at any time t is $a(t) = 2t - 7$. If the initial velocity of the particle is 6, at what time t during the interval $0 \leq t \leq 4$ is the particle farthest to the right?

- (A) 0
(B) 1
(C) 2
(D) 3
(E) 4



50. The equation $y = 2e^{6x} - 5$ is a particular solution to which of the following differential equations?

- (A) $y' - 6y - 30 = 0$
(B) $2y' - 12y + 5 = 0$
(C) $y'' - 5y' - 6y = 0$
(D) $y'' - 2y' + y + 5 = 0$

**Answer A**

Correct. One way to verify that a function is a solution to a differential equation is to check that the function and its derivatives satisfy the differential equation. The differential equation in this option involves y and y' . The correct derivative must be computed and the algebra correctly done to verify that the differential equation is satisfied.

$$y' = 12e^{6x}$$

$$y' - 6y - 30 = 12e^{6x} - 6(2e^{6x} - 5) - 30 = 12e^{6x} - 12e^{6x} + 30 - 30 = 0$$

4-2-2a

51.

t (minutes)	0	12	20	24	40
$v(t)$ (meters per minute)	0	200	240	-220	150

Johanna jogs along a straight path. For $0 \leq t \leq 40$, Johanna's velocity is given by a differentiable function v . Selected values of $v(t)$, where t is measured in minutes and $v(t)$ is measured in meters per minute, are given in the table above.

Bob is riding his bicycle along the same path. For $0 \leq t \leq 10$, Bob's velocity is modeled by $B(t) = t^3 - 6t^2 + 300$, where t is measured in minutes and $B(t)$ is measured in meters per minute. Find Bob's acceleration at time $t=5$.

Part C

1 point is earned for using $B'(t)$

1 point is earned for the answer

Bob's acceleration is $B'(t) = 3t^2 - 12t$.

$B'(5) = 3(25) - 12(5) = 15$ meters/min²



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for using $B'(t)$

1 point is earned for the answer

Bob's acceleration is $B'(t) = 3t^2 - 12t$.

$B'(5) = 3(25) - 12(5) = 15$ meters/min²

4-2-2a

52.

t (minutes)	0	12	20	24	40
$v(t)$ (meters per minute)	0	200	240	-220	150

Johanna jogs along a straight path. For $0 \leq t \leq 40$, Johanna's velocity is given by a differentiable function v . Selected values of $v(t)$, where t is measured in minutes and $v(t)$ is measured in meters per minute, are given in the table above.

(a) Use the data in the table to estimate the value of $v'(16)$.

(b) Using correct units, explain the meaning of the definite integral $\int_0^{40} |v(t)| dt$ in the context of the problem. Approximate the value of $\int_0^{40} |v(t)| dt$ using a right Riemann sum with the four subintervals indicated in the table.

(c) Bob is riding his bicycle along the same path. For $0 \leq t \leq 10$, Bob's velocity is modeled by $B(t) = t^3 - 6t^2 + 300$, where t is measured in minutes and $B(t)$ is measured in meters per minute. Find Bob's acceleration at time $t = 5$.

(d) Based on the model B from part (c), find Bob's average velocity during the interval $0 \leq t \leq 10$.

Part A

1 point is earned for the approximation

$$v'(16) \approx \frac{240-200}{20-12} = 5 \text{ meters/min}^2$$



0	1
---	---

The student response earns all of the following points:

1 point is earned for the approximation

$$v'(16) \approx \frac{240-200}{20-12} = 5 \text{ meters/min}^2$$

Part B

1 point is earned for the explanation

1 point is earned for the right Riemann sum

1 point is earned for the approximation

4-2-2a

$\int_0^{40} |v(t)| dt$ is the total distance Johanna jogs, in meters, over the time interval $0 \leq t \leq 40$ minutes.

$$\int_0^{40} |v(t)| dt \approx 12 \cdot |v(12)| + 8 \cdot |v(20)| + 4 \cdot |v(24)| + 16 \cdot |v(40)|$$

$$= 12 \cdot 200 + 8 \cdot 240 + 4 \cdot 220 + 16 \cdot 150$$

$$= 2400 + 1920 + 880 + 2400$$

$$= 7600 \text{ meters}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the explanation

1 point is earned for the right Riemann sum

1 point is earned for the approximation

$\int_0^{40} |v(t)| dt$ is the total distance Johanna jogs, in meters, over the time interval $0 \leq t \leq 40$ minutes.

$$\int_0^{40} |v(t)| dt \approx 12 \cdot |v(12)| + 8 \cdot |v(20)| + 4 \cdot |v(24)| + 16 \cdot |v(40)|$$

$$= 12 \cdot 200 + 8 \cdot 240 + 4 \cdot 220 + 16 \cdot 150$$

$$= 2400 + 1920 + 880 + 2400$$

$$= 7600 \text{ meters}$$

Part C

1 point is earned for using $B'(t)$

1 point is earned for the answer

Bob's acceleration is $B'(t) = 3t^2 - 12t$.

$$B'(5) = 3(25) - 12(5) = 15 \text{ meters/min}^2$$

4-2-2a



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for using $B'(t)$

1 point is earned for the answer

Bob's acceleration is $B'(t) = 3t^2 - 12t$.

$$B'(5) = 3(25) - 12(5) = 15 \text{ meters/min}^2$$

Part D

1 point is earned for the integral

1 point is earned for the antiderivative

1 point is earned for the answer

$$\begin{aligned} \text{Avg vel} &= \frac{1}{10} \int_0^{10} (t^3 - 6t^2 + 300) dt \\ &= \frac{1}{10} \left[\frac{t^4}{4} - 2t^3 + 300t \right]_0^{10} \\ &= \frac{1}{10} \left[\frac{10000}{4} - 2000 + 3000 \right] = 350 \text{ meters/min} \end{aligned}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the integral

1 point is earned for the antiderivative

1 point is earned for the answer

4-2-2a

$$\begin{aligned}\text{Avg vel} &= \frac{1}{10} \int_0^{10} (t^3 - 6t^2 + 300) dt \\ &= \frac{1}{10} \left[\frac{t^4}{4} - 2t^3 + 300t \right]_0^{10} \\ &= \frac{1}{10} \left[\frac{10000}{4} - 2000 + 3000 \right] = 350 \text{ meters/min}\end{aligned}$$

53.  The acceleration of a particle moving along the x -axis is given by $a(t) = (t - 8)\sin t$ for $0 \leq t \leq 8$. At what value of t is the particle's velocity decreasing most rapidly?

(A) 0

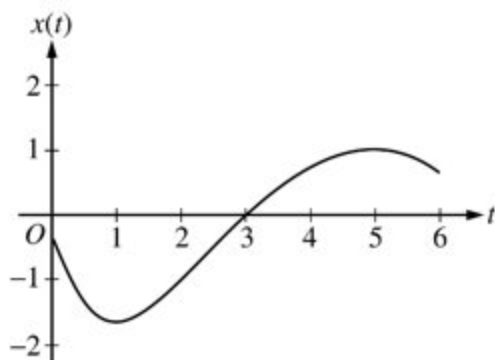
(B) 1.420

(C) 3.142

(D) 4.439



54.



A particle moves along a straight line. The graph of the particle's position $x(t)$ at time t is shown above for $0 < t < 6$. The graph has horizontal tangents at $t = 1$ and $t = 5$ and a point of inflection at $t = 2$. For what values of t is the velocity of the particle increasing?

(A) $0 < t < 2$ (B) $1 < t < 5$ (C) $2 < t < 6$ (D) $3 < t < 5$ only(E) $1 < t < 2$ and $5 < t < 6$ 

55.  The position of a particle moving along the x -axis is $x(t) = \sin(2t) - \cos(3t)$ for time $t \geq 0$. When $t = \pi$, the acceleration of the particle is

(A) 9

(B) $1/9$

(C) 0

(D) $-\frac{1}{9}$

(E) -9



4-2-2a

A particle moves along the x -axis so that at any time $t > 0$ its velocity is given by $v(t) = t \ln t - t$. At time $t = 1$, the position of the particle is $x(1) = 6$.

56. For what values of t is the particle moving to the right?

Part B

1 point is earned for correctly setting $t \ln t - t > 0$

$$v(t) = t \ln t - t > 0$$

$$t(\ln t - 1) > 0$$

1 point is earned for the correct answer

$$t > e$$



0	1	2
---	---	---

The student response earns two of the following points:

1 point is earned for correctly setting $t \ln t - t > 0$

$$v(t) = t \ln t - t > 0$$

$$t(\ln t - 1) > 0$$

1 point is earned for the correct answer

$$t > e$$

4-2-2a

57. Write an expression for the acceleration of the particle.
-

Part A

1 point is earned for the correct $v'(t)$

$$a(t) = v'(t) = \ln t + t \cdot \frac{1}{t} - 1 = \ln t$$



0

1

The student earns one of the following points:

1 point is earned for the correct $v'(t)$

$$a(t) = v'(t) = \ln t + t \cdot \frac{1}{t} - 1 = \ln t$$

4-2-2a

58. A particle moves along the x -axis so that at any time $t > 0$ its velocity is given by $v(t) = t \ln t - t$. At time $t = 1$, the position of the particle is $x(1) = 6$.
- Write an expression for the acceleration of the particle.
 - For what values of t is the particle moving to the right?
 - What is the minimum velocity of the particle? Show the analysis that leads to your conclusion.
 - Write an expression of the position $x(t)$ of the particle.

Part A

1 point is earned for the correct $v'(t)$

$$a(t) = v'(t) = \ln t + t \cdot \frac{1}{t} - 1 = \ln t$$



0	1
---	---

The student earns all of the following points:

1 point is earned for the correct $v'(t)$

$$a(t) = v'(t) = \ln t + t \cdot \frac{1}{t} - 1 = \ln t$$

Part B

1 point is earned for correctly setting $t \ln t - t > 0$

$$v(t) = t \ln t - t > 0$$

$$= t (\ln t - 1) > 0$$

1 point is earned for the correct answer

$$t > e$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for correctly setting $t \ln t - t > 0$

$$v(t) = t \ln t - t > 0$$

4-2-2a

$$= t(\ln t - 1) > 0$$

1 point is earned for the correct answer

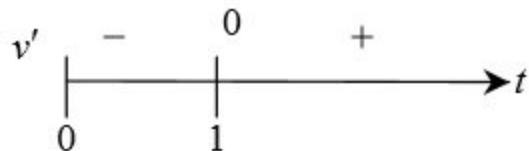
$$t > e$$

Part C1 point is earned for correctly solving $v'(t) = 0$

$$v'(t) = \ln t = 0$$

$$t = 1$$

1 point is earned for the correct analysis



1 point is earned for the correct answer

minimum velocity is $v(1) = -1$ Max 1/3 if student's v' does not involve $\ln$ 

0	1	2	3
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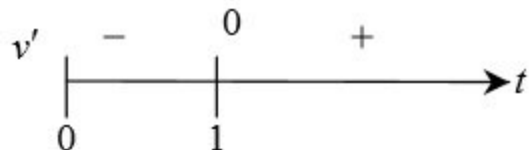
The student response earns all of the following points:

1 point is earned for correctly solving $v'(t) = 0$

$$v'(t) = \ln t = 0$$

$$t = 1$$

1 point is earned for the correct analysis



1 point is earned for the correct answer

4-2-2a

minimum velocity is $v(1) = -1$

Max 1/3 if student's v' does not involve $\ln$

Part D

2 points are earned for the correct antiderivative of $t \ln t - t$

<-2>: error in $\int$ -by-parts

<-1>: any other error

$$\begin{aligned} \int t \ln t - t \, dt \\ &= \left(\frac{1}{2}t^2 \ln t - \frac{1}{4}t^2 \right) - \frac{1}{2}t^2 + C \\ &= \frac{1}{2}t^2 \ln t - \frac{3}{4}t^2 + C; \\ 6 &= x(1) = 0 - \frac{3}{4} + C \end{aligned}$$

1 point is earned for correctly solving for C

$$\begin{aligned} C &= \frac{27}{4}; \\ x(t) &= \frac{1}{2}t^2 \ln t - \frac{3}{4}t^2 + \frac{27}{4} \end{aligned}$$

0/3 if not antidifferentiating $t \ln t - t$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for the correct antiderivative of $t \ln t - t$

<-2>: error in $\int$ -by-parts

<-1>: any other error

4-2-2a

$$\begin{aligned}& \int t \ln t - t \, dt \\&= \left(\frac{1}{2} t^2 \ln t - \frac{1}{4} t^2 \right) - \frac{1}{2} t^2 + C \\&= \frac{1}{2} t^2 \ln t - \frac{3}{4} t^2 + C; \\&6 = x(1) = 0 - \frac{3}{4} + C\end{aligned}$$

1 point is earned for correctly solving for C

$$\begin{aligned}C &= \frac{27}{4}; \\x(t) &= \frac{1}{2} t^2 \ln t - \frac{3}{4} t^2 + \frac{27}{4}\end{aligned}$$

0/3 if not antidifferentiating $t \ln t - t$

4-2-2a

59. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

A particle moves along the x -axis so that its velocity at time t is given by $v(t) = \frac{t^6 - 13t^4 + 12}{10t^3 + 3}$. At time $t = 0$, the initial position of the particle is $x = 7$.



(a) Find the acceleration of the particle at time $t = 5.1$.

(b) Find all values of t in the interval $0 \leq t \leq 2$ for which the speed of the particle is 1.

(c) Find the position of the particle at time $t = 4$. Is the particle moving toward the origin or away from the origin at time $t = 4$? Justify your answer.

(d) During the time interval $0 < t \leq 4$, does the particle return to its initial position? Give a reason for your answer.

Part A

1 point is earned for: answer

$$v'(5.1) = 6.491816$$

The acceleration of the particle at time $t = 5.1$ is 6.492 (or 6.491).



0	1
---	---

The student response earns all of the following points:

1 point is earned for: answer

$$v'(5.1) = 6.491816$$

The acceleration of the particle at time $t = 5.1$ is 6.492 (or 6.491).

Part B

1 point is earned for: considers $|v(t)| = 1$

1 point is earned for: answer

$$|v(t)| = 1 \text{ for } 0 \leq t \leq 2 \Rightarrow t = 0.771829, t = 1.400556$$

The speed of the particle is 1 at times $t = 0.772$ (or 0.771) and $t = 1.401$ (or 1.400).



0	1	2
---	---	---

4-2-2a

The student response earns all of the following points:

1 point is earned for: considers $|v(t)| = 1$

1 point is earned for: answer

$$|v(t)| = 1 \text{ for } 0 \leq t \leq 2 \Rightarrow t = 0.771829, t = 1.400556$$

The speed of the particle is 1 at times $t = 0.772$ (or 0.771) and $t = 1.401$ (or 1.400).

Part C

1 point is earned for: integral

1 point is earned for: uses initial condition

1 point is earned for: position

1 point is earned for: movement relative to the origin, with justification

Let $x(t)$ be the position of the particle at time t .

$$x(4) = 7 + \int_0^4 v(t) dt = 6.711558$$

The position of the particle at time $t = 4$ is 6.712 (or 6.711).

The particle is moving away from the origin because $x(4) > 0$ and $v(4) = 1.213064 > 0$.



0	1	2	3	4
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The student response earns all of the following points:

1 point is earned for: integral

1 point is earned for: uses initial condition

1 point is earned for: position

1 point is earned for: movement relative to the origin, with justification

Let $x(t)$ be the position of the particle at time t .

$$x(4) = 7 + \int_0^4 v(t) dt = 6.711558$$

4-2-2a

The position of the particle at time $t = 4$ is 6.712 (or 6.711).

The particle is moving away from the origin because $x(4) > 0$ and $v(4) = 1.213064 > 0$.

part D

1 point is earned for: sandwiches initial position

1 point is earned for: answer with reason

$x'(t) = v(t) \Rightarrow x(t)$ is continuous on $0 < t \leq 4$.

$$x(1) = 7 + \int_0^1 v(t) dt = 9.403593$$

From part (c), $x(4) = 6.712$.

From time $t = 1$ to $t = 4$, the particle moves from $x = 9.404$ to $x = 6.712$. By the Intermediate Value Theorem, the particle must return to $x = 7$ during the time interval.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for: sandwiches initial position

1 point is earned for: answer with reason

$x'(t) = v(t) \Rightarrow x(t)$ is continuous on $0 < t \leq 4$.

$$x(1) = 7 + \int_0^1 v(t) dt = 9.403593$$

From part (c), $x(4) = 6.712$.

From time $t = 1$ to $t = 4$, the particle moves from $x = 9.404$ to $x = 6.712$. By the Intermediate Value Theorem, the particle must return to $x = 7$ during the time interval.

4-2-2a

60.  A particle moves along the x -axis so that at any time $t \geq 0$ its velocity is given by $v(t) = t^2 \ln(t + 2)$. What is the acceleration of the particle at time $t = 6$?

- (A) 1.500
(B) 20.453
(C) 29.453
(D) 74.860
(E) 133.417



4-2-2a

61. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

A particle moves along the y -axis so that its position at time t is given by $y(t) = t^2 \tan\left(\frac{1}{t}\right)$ for $t > 1$.

(a) Show that the velocity of the particle at time t is given by $v(t) = 2t \tan\left(\frac{1}{t}\right) - \sec^2\left(\frac{1}{t}\right)$ for $t > 1$.

(b) At time $t = \frac{4}{\pi}$, is the particle moving toward the origin or away from the origin? Give a reason for your answer.

(c) The velocity of the particle at time t can be written as $v(t) = \frac{2 \tan\left(\frac{1}{t}\right)}{\frac{1}{t}} - \sec^2\left(\frac{1}{t}\right)$ for $t > 1$.

Find $\lim_{t \rightarrow \infty} v(t)$. Show the work that leads to your answer.

Part A

The first and second points require evidence of product rule and chain rule and no errors. At most 1 out of 2 points is earned for partial communication of product rule and chain rule with a maximum of one computational error.

The third point is earned for a response that arrives at the given expression rather than an algebraically equivalent expression.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ product rule
- ☐ chain rule
- ☐ verification

Solution:

4-2-2a

$$v(t) = y'(t) = 2t \tan\left(\frac{1}{t}\right) + t^2 \sec^2\left(\frac{1}{t}\right) \left(-\frac{1}{t^2}\right) = 2t \tan\left(\frac{1}{t}\right) - \sec^2\left(\frac{1}{t}\right)$$

Part B

For the first and second points, trigonometric function values do not need to be evaluated; however, the sign of the expression must be determined correctly to earn the point.

The response is eligible for the third point if either of the first 2 points is earned and the answer with reason is consistent with the previous results.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ $y\left(\frac{4}{\pi}\right) > 0$
- ☐ $v\left(\frac{4}{\pi}\right) > 0$
- ☐ answer with reason

Solution:

$$y\left(\frac{4}{\pi}\right) = \left(\frac{4}{\pi}\right)^2 \tan\left(\frac{\pi}{4}\right) = \frac{16}{\pi^2} > 0$$

$$v\left(\frac{4}{\pi}\right) = 2\left(\frac{4}{\pi}\right) \tan\left(\frac{\pi}{4}\right) - \sec^2\left(\frac{\pi}{4}\right) = \frac{8}{\pi} - 2 > 0$$

Because $y\left(\frac{4}{\pi}\right) > 0$, the particle is above the origin at time $t = \frac{4}{\pi}$.

Because $v\left(\frac{4}{\pi}\right) > 0$, the particle is moving up at time $t = \frac{4}{\pi}$.

Therefore, the particle is moving away from the origin at time $t = \frac{4}{\pi}$.

Part C

The first point does not require supporting work.

The second point requires showing that an indeterminate form $\frac{0}{0}$ is present in order to apply L'Hospital's Rule. Any error in mathematical communication (e.g., writing $= \frac{0}{0}$ or not using limit notation) impacts the second point.

The third point does not require a simplified answer. Any differentiation or computation error impacts the third point.

4-2-2a

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ $\lim_{t \rightarrow \infty} \sec^2 \left(\frac{1}{t} \right) = 1$
- ☐ application of L'Hospital's Rule
- ☐ answer

Solution:

$$\lim_{t \rightarrow \infty} 2 \tan \left(\frac{1}{t} \right) = 2 \tan 0 = 0$$

$$\lim_{t \rightarrow \infty} \frac{1}{t} = 0$$

$$\lim_{t \rightarrow \infty} \sec^2 \left(\frac{1}{t} \right) = \sec^2 0 = 1$$

By L'Hospital's Rule,

$$\lim_{t \rightarrow \infty} \frac{2 \tan \left(\frac{1}{t} \right)}{\frac{1}{t}} = \lim_{t \rightarrow \infty} \frac{2 \sec^2 \left(\frac{1}{t} \right) \left(-\frac{1}{t^2} \right)}{-\frac{1}{t^2}} = \lim_{t \rightarrow \infty} 2 \sec^2 \left(\frac{1}{t} \right) = 2$$

Therefore,

$$\lim_{t \rightarrow \infty} v(t) = \lim_{t \rightarrow \infty} \left(\frac{2 \tan \left(\frac{1}{t} \right)}{\frac{1}{t}} - \sec^2 \left(\frac{1}{t} \right) \right) = 2 - 1 = 1$$

4-2-2a

62. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

The position of a particle moving along the x -axis at time t is given by the twice-differentiable function $x(t)$, where $x(0) = 2$. For $0 \leq t \leq 5$, the derivative of $x(t)$ is given by $x'(t) = \sin(t^{1.1})\sqrt{10t+4}$.

- (a) Find all times t in the open interval $0 < t < 5$ when the particle changes direction. Give a reason for your answer.
- (b) Find the total distance traveled by the particle from time $t = 1$ to time $t = 4$.
- (c) Find the acceleration of the particle at time $t = 4$. Is the speed of the particle increasing or decreasing at time $t = 4$? Give a reason for your answer.
- (d) Find the position of the particle at time $t = 5$. For $0 \leq t \leq 5$, is there a time t when the particle is at the origin? Justify your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point cannot be earned implicitly by reporting $t = 2.831095$; the setup must be shown.

A response that finds an incorrect value of t does not earn the second point.

A response may find and report extraneous solutions to the equation $x'(t) = 0$. If the response does not specifically eliminate the extraneous solutions, it will not earn the second point.

The second point can be earned by applying the Second Derivative Test. This includes reporting that $x'(t) = 0$ at $t = 2.831$, and that $x''(2.831) < 0$, and therefore the particle changes direction when $t = 2.831$. Such a response need not report the value of $x''(2.831)$, but if a value is presented it must be correct for the presented digits, up to three decimal places ($x''(2.831) = -6.938$).

4-2-2a

Degree mode: A response that presents answers obtained by using a calculator in degree mode does not earn the first point it would have otherwise earned. The response is eligible for all subsequent points (unless no answer is possible in degree mode or the question is made simpler by using degree mode). In degree mode, $t = 112.26587$, which is not in $0 < t < 5$. Therefore, such a response does not earn the second point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Sets $x'(t) = 0$
- ☐ Answer with reason

Solution:

$$x'(t) = 0 \Rightarrow t = 2.831095$$

The particle changes direction when $t = 2.831$ because the velocity, $x'(t)$, changes from positive to negative at this value.

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response that presents incorrect limits or no limits does not earn the first point, but, in the presence of the correct answer, does earn the second point.

A response that omits the absolute value on the integrand or uses $x(t)$ for the integrand does not earn the first point, but, in the presence of the correct answer, does earn the second point.

A response that presents $\left| \int_1^4 x'(t) dt \right|$ does not earn the first point, but, in the presence of the correct answer, does earn the second point.

Degree mode: In degree mode, $\int_1^4 |x'(t)| dt = 0.813$ (or 0.812). (See degree mode statement in part (a).)



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Integral(s)

4-2-2a

☐ Answer
Solution:

$$\int_1^4 |x'(t)| dt = 10.5556$$

The total distance traveled by the particle from time $t = 1$ to time $t = 4$ is 10.556 (or 10.555).

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The minimum work needed to earn the first point is $x''(4) = -1.732$ (or -1.731).

$a(4) = -1.732$ (or -1.731) is not sufficient to earn the first point. The connection $a(4) = x''(4)$, or $a(t) = x''(t)$, must be explicitly made.

A response does not need to find the value of $v(4)$; the sign is sufficient. However, a response that reports an incorrect sign or value of $v(4)$ does not earn the second point. A response must declare a value for $x''(4)$ to be eligible for the second point.

If a value for $v(4)$ is presented, that value must be accurate for the digits shown, up to three decimal places.

A response that declares an incorrect value of $x''(4)$ is eligible for the second point. The conclusion must be consistent with the presented value of $x''(4)$ and the correct sign/value of $v(4)$.

Degree mode: In degree mode, $x''(4) = a(4) = 0.206$ (or 8.415) and $x'(4) = v(4) = 0.531 > 0$. (See degree mode statement in part (a).)



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ $x''(4)$
- ☐ Conclusion with reason

Solution:

$$x''(4) = a(4) = -1.732 \text{ (or } -1.731)$$

$$x'(4) = v(4) = -6.587438$$

4-2-2a

Because the values of the velocity and acceleration have the same sign at $t = 4$, the speed is increasing at time $t = 4$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response that presents an integral of $\int_0^5 |x'(t)| dt$ does not earn the first point.

· This response will earn the second point only for a value of $x(5) = 2 + \int_0^5 |x'(t)| dt = 19.612$.

· If this response earns the second point, it will earn the third point only for reasoning that there is not sufficient information to determine whether there is a time t , for $0 \leq t \leq 5$, when the particle is at the origin. The reasoning must reference the Intermediate Value Theorem.

To earn the third point, the response must include:

- differentiability implies continuity (the continuity of $x(t)$ by itself is insufficient),
- $x(5) < 0 < x(0)$ or -0.566 (or -0.565) $< 0 < 2$ (this may be expressed in words or symbols), and
- “yes.”

A response of $x(5) = \int_0^5 x'(t) dt = -2.566$ (or -2.565) earns the first point, does not earn the second point, and is eligible to earn the third point.

A response with incorrect or no limits of integration does not earn the first point. Such a response that presents the correct value of $x(5)$ does earn the second point, and is eligible to earn the third point. A response with incorrect or no limits that presents an incorrect value of $x(5)$ will not earn any points.

A response that solves the equation $2 + \int_0^t x'(s) ds = 0$ for t and reports a value of $t = 4.841$ earns the third point, but must correctly find $x(5)$ in order to earn the first 2 points.

Degree mode: In degree mode, $x(5) = 2 + \int_0^5 x'(t) dt = 3.479$. (See degree mode statement in part (a).)



0	1	2	3
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The student response accurately includes **one** of the criteria below.

☐ $\int_0^5 x'(t) dt$

4-2-2a☐ $x(5)$ ☐ Answer with justification

$$x(5) = 2 + \int_0^5 x'(t) dt = -0.565953$$

The position of the particle at time $t = 5$ is -0.566 (or -0.565).

The particle is at the origin when $x(t) = 0$.

$x(t)$ is twice-differentiable. $\Rightarrow x(t)$ is continuous.

Because $x(5) < 0 < x(0) = 2$ and $x(t)$ is continuous, by the Intermediate Value Theorem there must be a time t in $0 \leq t \leq 5$ when the particle is at the origin.

4-2-2a

63. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

A particle, P , is moving along the x -axis. The velocity of particle P at time t is given by $v_P(t) = \sin(t^{1.5})$ for $0 \leq t \leq \pi$. At time $t = 0$, particle P is at position $x = 5$.

A second particle, Q , also moves along the x -axis. The velocity of particle Q at time t is given by $v_Q(t) = (t - 1.8) \cdot 1.25^t$ for $0 \leq t \leq \pi$. At time $t = 0$, particle Q is at position $x = 10$.

(a) Find the positions of particles P and Q at time $t = 1$.

(b) Are particles P and Q moving toward each other or away from each other at time $t = 1$? Explain your reasoning.

(c) Find the acceleration of particle Q at time $t = 1$. Is the speed of particle Q increasing or decreasing at time $t = 1$? Explain your reasoning.

(d) Find the total distance traveled by particle P over the time interval $0 \leq t \leq \pi$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for the explicit presentation of at least one definite integral, either $\int_0^1 v_P(t) dt$ or $\int_0^1 v_Q(t) dt$.

The first point must be earned to be eligible for the second and third points.

The second point is earned for adding the initial condition to at least one of the definite integrals and finding the correct position.

4-2-2a

Writing $\int_0^1 v_P(t) + 5 = 5.370660$ does not earn a position point, because the missing dt makes this statement unclear or false. However, $5 + \int_0^1 v_P(t) = 5.370660$ does earn the position point because it is not ambiguous. Similarly, for the position of Q .

Read unlabeled answers presented left to right, or top to bottom, as $x_P(1)$ and $x_Q(1)$, respectively.

Special case 1: A response of $x_P(1) = 5 + \int_0^a v_P(t)dt = 5.370660$ AND

$x_Q(1) = 10 + \int_0^a v_Q(t)dt = 8.564355$ for $a \neq 1$ earns one point.

Special case 2: A response of $x_P(1) = 5 + \int v_P(t)dt = 5.370660$ AND $x_Q(1) = 10 + \int v_Q(t)dt = 8.564355$ or the equivalent, never providing the definite integrals, earns one point.

Degree mode: A response that presents answers obtained by using a calculator in degree mode does not earn the first point it would have otherwise earned. The response is generally eligible for all subsequent points (unless no answer is possible in degree mode or the question is made simpler by using degree mode). In degree mode, $x_P(1)$ is 5.007 (or 5.006).



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ One definite integral
- ☐ One position
- ☐ The other position

Solution:

$$x_P(1) = 5 + \int_0^1 v_P(t)dt = 5.370660$$

At time $t = 1$, the position of particle P is $x = 5.371$ (or 5.370).

$$x_Q(1) = 10 + \int_0^1 v_Q(t)dt = 8.564355$$

At time $t = 1$, the position of particle Q is $x = 8.564$.

Part B

4-2-2a

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for using the sign of $v_P(1)$ or $v_Q(1)$ to determine the direction of motion for one of the particles. This point cannot be earned without reference to the sign of $v_P(1)$ or $v_Q(1)$.

It is not necessary to present an explicit value for $v_P(1)$ or $v_Q(1)$, but if a value is presented, it must be correct as far as reported, up to three places after the decimal.

Read with imported incorrect position values from part (a).

If one or both position values were not found in part (a), but are found in part (b), the points for part (a) are not earned retroactively.

To earn the second point the explanation must be based on the signs of $v_P(1)$ and $v_Q(1)$ and the relative positions of particle P and particle Q at $t = 1$. References to other values of time, such as $t = 0$, are not sufficient.

Degree mode: $v_P(1) = 0.017$. (See degree mode statement in part (a).)



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Direction of motion for one particle
- ☐ Answer with explanation

Solution:

$$v_P(1) = \sin(1^{1.5}) = 0.841471 > 0$$

At time $t = 1$, particle P is moving to the right.

$$v_Q(1) = (1 - 1.8) \cdot 1.25^1 = -1 < 0$$

At time $t = 1$, particle Q is moving to the left.

At time $t = 1$, $x_P(1) < x_Q(1)$, so particle P is to the left of particle Q .

Thus, at time $t = 1$, particles P and Q are moving toward each other.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point the acceleration must be explicitly connected to v'_Q (e.g., $v'_Q(1) = 1.026856$).

4-2-2a

The first point is not earned for an unsupported value of 1.027 (or 1.026). The setup, $v'_Q(1)$, must be shown. Presenting only $a_Q(1) = 1.027$ (or 1.026) without indication that $v'_Q = a_Q$ is not enough to earn the first point.

A response does not need to present a value for $v_Q(1)$; the sign is sufficient.

To earn the second point a response must compare the signs of a_Q and v_Q at $t = 1$. Considering only one sign is not sufficient.

After the first point has been earned, a response declaring only “velocity and acceleration are of opposite signs at $t = 1$ so the particle is slowing down” (or equivalent) earns the second point.

The second point may be earned without the first, as long as the response does not present an incorrect value or sign for $v_Q(1)$ and concludes the particle is slowing down because velocity and acceleration have opposite signs at $t = 1$.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Setup and acceleration
- ☐ Speed decreasing with reason

Solution:

$$a_Q(1) = v'_Q(1) = 1.026856$$

The acceleration of particle Q is 1.027 (or 1.026) at time $t = 1$.

$$v_Q(1) = -1 < 0 \text{ and } a_Q(1) > 0$$

The speed of particle Q is decreasing at time $t = 1$ because the velocity and acceleration have opposite signs.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for $\int_0^\pi |v_P(t)| dt$.

The first point can also be earned for a sum (or difference) of definite integrals, such as a

$$\int_0^{2.145029} v_P(t) dt - \int_{2.145029}^\pi v_P(t) dt, \text{ provided the response has indicated } v_P(2.145029) = 0.$$

The second point can only be earned for the correct answer.

4-2-2a

The unsupported value 1.931 earns no points.

A response reporting the distance traveled by particle Q as $\int_0^\pi |v_Q(t)|dt = 3.506$ earns the first point and is not eligible for the second point.

In degree mode, the total distance traveled is 0.122. (See degree mode statement in part (a).) In the degree mode case, the response must present $\int_0^\pi |v_P(t)|dt$ in order to earn the first point because

$$\int_0^\pi |v_P(t)|dt = \int_0^\pi v_P(t)dt.$$



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Definite integral
- ☐ Answer

Solution:

$$\int_0^\pi |v_P(t)|dt = 1.93148$$

Over the time interval $0 \leq t \leq \pi$, the total distance traveled by particle P is 1.931.

64. A particle moves along a line so that its velocity is given by $v(t) = -t^3 + 2t^2 + 2^{-t}$ for $t \geq 0$. For what values of t is the speed of the particle increasing?

- (A) $(0, 0.177)$ and $(1.256, \infty)$
- (B) $(0, 1.256)$ only
- (C) $(0, 2.057)$ only
- (D) $(0.177, 1.256)$ only
- (E) $(0.177, 1.256)$ and $(2.057, \infty)$



65. A particle moves along the x -axis so that at time $t \geq 0$ its position is given by $x(t) = \cos \sqrt{t}$. What is the velocity of the particle at the first instance the particle is at the origin?

- (A) -1
- (B) -0.624
- (C) -0.318
- (D) 0
- (E) 0.065



4-2-2a

66. For $t \geq 0$, the position of a particle moving along the x -axis is given by $x(t) = \sin t - \cos t$. What is the acceleration of the particle at the point where the velocity is first equal to 0 ?

(A) $-\sqrt{2}$



(B) -1

(C) 0

(D) 1

(E) $\sqrt{2}$

4-2-2a

67. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

A particle moves along the x -axis so that its position at time $t > 0$ is given by $x(t) = \frac{t^2 - 9}{3t^2 + 8}$.

- Show that the velocity of the particle at time t is given by $v(t) = \frac{70t}{(3t^2 + 8)^2}$.
- Is the particle moving toward the origin or away from the origin at time $t = 2$? Give a reason for your answer.
- The acceleration of the particle is given by $a(t)$. Write an expression for $a(t)$, and find the value of $a(2)$.
- What position does the particle approach as t approaches infinity?

Part A

The first point is earned for successful demonstration of the quotient rule.

The second point is earned for a response that arrives at the given expression rather than an algebraically equivalent expression.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ quotient rule
- ☐ verification

Solution:

4-2-2a

$$v(t) = x'(t) = \frac{2t(3t^2+8) - (t^2-9)6t}{(3t^2+8)^2}$$

$$= \frac{6t^3+16t-6t^3+54t}{(3t^2+8)^2} = \frac{70t}{(3t^2+8)^2}$$

Part B

The response is eligible for the third point if either of the first 2 points is earned and the answer with reason is consistent with the previous results.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ $x(2) < 0$
- ☐ $v(2) > 0$
- ☐ answer with reason

Solution:

$$x(2) = \frac{-5}{20} = -\frac{1}{4} < 0$$

$$v(2) = \frac{140}{20^2} = \frac{7}{20} > 0$$

Because $x(2) < 0$, the particle is to the left of the origin at time $t = 2$.

Because $v(2) > 0$, the particle is moving to the right at time $t = 2$.

Therefore, the particle is moving toward the origin at time $t = 2$.

Part C

The first and second points require evidence of quotient rule and chain rule and no errors. At most 1 out of 3 points is earned for partial communication of quotient rule and chain rule with a maximum of one computational error.

A simplified answer is not required for the third point.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

4-2-2a



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ quotient rule
- ☐ chain rule
- ☐ answer

Solution:

$$a(t) = v'(t) = \frac{70(3t^2+8)^2 - 70t \cdot 2(3t^2+8) \cdot 6t}{(3t^2+8)^4}$$

$$= \frac{70(3t^2+8) - 70 \cdot 12t^2}{(3t^2+8)^3} = \frac{70(8-9t^2)}{(3t^2+8)^3}$$

$$a(2) = \frac{70(8-9(2)^2)}{(3(2)^2+8)^3} = \frac{70(-28)}{20^3} = -\frac{49}{200}$$

Part D

Supporting work is not required to earn the point.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The student response accurately includes a correct position.

Solution:

$$\lim_{t \rightarrow \infty} x(t) = \lim_{t \rightarrow \infty} \frac{t^2-9}{3t^2+8} = \frac{1}{3}$$

The particle approaches $x = \frac{1}{3}$ as t approaches infinity.

4-2-2a

68. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Particle P moves along the x -axis so that its position at time $t > 0$ is given by $x_P(t) = \frac{e^{2-t}-2t}{e^{2-t}+3t}$. A second particle, particle Q , also moves along the x -axis so that its position at time t is given by $x_Q(t) = t^3 - 3t^2 + 5$.

- Show that the velocity of particle P at time t is given by $v_P(t) = \frac{-5te^{2-t}-5e^{2-t}}{(e^{2-t}+3t)^2}$.
- At time $t = 2$, particle Q is at rest. At time $t = 2$, is particle P moving toward particle Q or away from particle Q ? Explain your reasoning.
- The acceleration of particle Q is given by $a_Q(t)$. Find the value of $a_Q(2)$.
- Describe the position of particle P and the position of particle Q as t approaches infinity. Show the work that leads to your answers.

Part A

The first point is earned for successful demonstration of the quotient rule.

The second point is earned for a response that arrives at the given expression rather than an algebraically equivalent expression.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ quotient rule
- ☐ verification

Solution:

4-2-2a

$$v_P(t) = x'(t) = \frac{(-e^{2-t}-2)(e^{2-t}+3t) - (e^{2-t}-2t)(-e^{2-t}+3)}{(e^{2-t}+3t)^2}$$

$$= \frac{(-e^{4-2t}-3te^{2-t}-2e^{2-t}-6t) - (-e^{4-2t}+3e^{2-t}+2te^{2-t}-6t)}{(e^{2-t}+3t)^2} = \frac{-5te^{2-t}-5e^{2-t}}{(e^{2-t}+3t)^2}$$

Part B

The response is eligible for the third point if either of the first 2 points is earned and the answer with reason is consistent with the previous results.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ $x_P(2)$ and $x_Q(2)$
- ☐ $v_P(2) < 0$
- ☐ answer with explanation

Solution:

$$x_P(2) = \frac{1-4}{1+6} = -\frac{3}{7};$$

$$x_Q(2) = 1$$

$$v_P(2) = \frac{-10-5}{7^2} = -\frac{15}{49} < 0$$

Because $x_P(2) < x_Q(2)$, particle P is to the left of particle Q at time $t = 2$.

Because $v_P(2) < 0$, P is moving to the left at time $t = 2$.

Because particle Q is at rest at time $t = 2$, particle P is moving away from particle Q at time $t = 2$.

Part C

Supporting work is not required to earn the point.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

4-2-2a



0	1
---	---

The student response accurately includes a correct value of $a_Q(2)$

$$v_Q(t) = x_Q'(t) = 3t^2 - 6t$$

$$a_Q(t) = v_Q'(t) = 6t - 6$$

$$a_Q(2) = 12 - 6 = 6$$

Part D

The first point requires showing that an indeterminate form $\frac{\infty}{\infty}$ is present in order to apply L'Hospital's Rule. Any error in mathematical communication (e.g., writing " $\frac{\infty}{\infty}$ " or not using limit notation) impacts the first point.

For the second and third points, at most 1 out of 2 points may be earned for both correct limits without a connection to the interpretation of position.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ applies L'Hospital's Rule
- ☐ description of position of particle P
- ☐ description of position of particle Q

Solution:

$$\lim_{t \rightarrow \infty} e^{2-t} - 2t = -\infty$$

$$\lim_{t \rightarrow \infty} e^{2-t} + 3t = \infty$$

By L'Hospital's Rule,

$$\lim_{t \rightarrow \infty} x_P(t) = \lim_{t \rightarrow \infty} \frac{e^{2-t} - 2t}{e^{2-t} + 3t} = \lim_{t \rightarrow \infty} \frac{-e^{2-t} - 2}{-e^{2-t} + 3} = -\frac{2}{3}$$

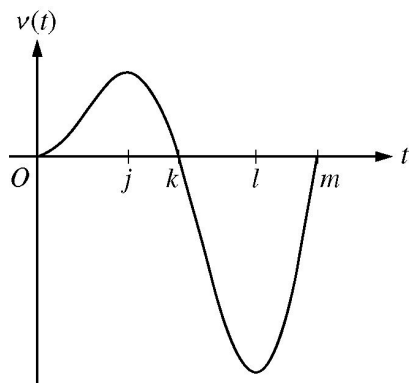
Therefore, particle P approaches $x = -\frac{2}{3}$ as t approaches infinity.

4-2-2a

$$\lim_{t \rightarrow \infty} x_Q(t) = \lim_{t \rightarrow \infty} (t^3 - 3t^2 + 5) = \infty$$

Therefore, the position of particle Q grows without bound as t approaches infinity.

69.



A particle moves along a straight line. The graph of the particle's velocity $v(t)$ at time t is shown above for $0 \leq t \leq m$, where j , k , l , and m are constants. The graph intersects the horizontal axis at $t = 0$, $t = k$, and $t = m$ and has horizontal tangents at $t = j$ and $t = l$. For what values of t is the speed of the particle decreasing?

- (A) $j \leq t \leq l$
 (B) $k \leq t \leq m$
 (C) $j \leq t \leq k$ and $l \leq t \leq m$ ✓
 (D) $0 \leq t \leq j$ and $k \leq t \leq l$
 (E) $0 \leq t \leq j$ and $l \leq t \leq m$

70. The position of a particle moving along a line is given by $s(t) = 2t^3 - 24t^2 + 90t + 7$ for $t \geq 0$. For what values of t is the speed of the particle increasing?

- (A) $3 < t < 4$ only
 (B) $t > 4$ only
 (C) $t > 5$ only
 (D) $0 < t < 3$ and $t > 5$
 (E) $3 < t < 4$ and $t > 5$ ✓

71.  A particle moves along a straight line with velocity given by $v(t) = 7 - (1.01)^{-t^2}$ at time $t \geq 0$. What is the acceleration of the particle at time $t = 3$?

- (A) -0.914
 (B) 0.055 ✓
 (C) 5.486
 (D) 6.086
 (E) 18.087

4-2-2a

72. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Particle P moves along the x -axis such that, for time $t > 0$, its position is given by $x_P(t) = 6 - 4e^{-t}$.

Particle Q moves along the y -axis such that, for time $t > 0$, its velocity is given by $v_Q(t) = \frac{1}{t^2}$. At time $t = 1$, the position of particle Q is $y_Q(1) = 2$.

- (a) Find $v_P(t)$, the velocity of particle P at time t .
- (b) Find $a_Q(t)$, the acceleration of particle Q at time t . Find all times t , for $t > 0$, when the speed of particle Q is decreasing. Justify your answer.
- (c) Find $y_Q(t)$, the position of particle Q at time t .
- (d) As $t \rightarrow \infty$, which particle will eventually be farther from the origin? Give a reason for your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response that equates $x_P(t)$ with $v_P(t)$ does not earn the point.

An unlabeled response earns the point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$v_P(t) = x_P'(t) = 4e^{-t}$$

Part B

4-2-2a

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earning the first point is not necessary for a response to be eligible to earn the second or third points; however, the response must present an expression for $a_Q(t)$ to be eligible for third point.

A response earns the second point with either of the following statements: “ $v_Q(t)$ and $a_Q(t)$ have opposite signs” or “ $v_Q(t)$ and $a_Q(t)$ have the same sign.” This statement, however, must be consistent with $v_Q(t)$ and the presented expression for $a_Q(t)$.

A response must earn the second point to be eligible for the third point. The answer must be consistent with the presented justification. Furthermore, responses for which $a_Q(t) > 0$ for $t > 0$ must conclude that there is no time at which the speed of the particle is decreasing.

A response that indicates $v_Q(t) < 0$ does not earn the third point, even if the answer and justification are consistent with a reported sign of $a_Q(t)$.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ $a_Q(t)$
- ☐ Considers signs of $a_Q(t)$ and $v_Q(t)$
- ☐ Answer with justification

Solution:

$$a_Q(t) = v_Q'(t) = \frac{-2}{t^3}$$

For $t > 0$, $a_Q(t) < 0$ and $v_Q(t) > 0$.

Because the velocity and acceleration have opposite signs, the speed of particle Q is decreasing for all $t > 0$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response that presents $\int_1^t \frac{1}{t^2} dt$ (using the same variable as a limit and integrand function) does not earn the first point unless it is followed by an attempt at integration.

A response that presents either $\int \frac{1}{t^2} dt$ or $-\frac{1}{t}$ (with no integral) earns the first point. If the response continues and presents $2 = -1 + C$, then the response earns the second point.

4-2-2a

A response that presents only $y_Q(t) = -\frac{1}{t} + 3$ will earn all 3 points. Note that the right side of this equation suffices to earn all points. A response of $y_Q(t) = -\frac{1}{t} + C$, where $C \neq 3$, (with no additional supporting work) earns only the first point.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Integral
- ☐ Uses initial condition
- ☐ Answer

Solution:

$$y_Q(t) = y_Q(1) + \int_1^t \frac{1}{s^2} ds$$

$$= 2 - \left(\frac{1}{s} \Big|_1^t \right) = 2 - \frac{1}{t} + 1 = 3 - \frac{1}{t}$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response with an incorrect $y_Q(t)$ from part (c) is eligible for both points in part (d) provided $y_Q(t)$ is a non-constant function. The second point is earned for a consistent answer with reason, and limits correct for particle P and the presented $y_Q(t)$.

Responses that present statements such as “ $6 - 4e^{-t}$ approaches 6” or “ Q goes to 3” earn the first point and are eligible for the second point.

A response that treats ∞ as an input for $x_P(t)$ or $y_Q(t)$, such as “ $6 - 4e^{-\infty}$ ” or “ $3 - \frac{1}{\infty}$ ” is not eligible for the second point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ One correct limit
- ☐ Answer with reason

4-2-2a**Solution:**

For particle P , $\lim_{t \rightarrow \infty} (6 - 4e^{-t}) = 6$.

For particle Q , $\lim_{t \rightarrow \infty} (3 - \frac{1}{t}) = 3$.

Because $6 > 3$, particle P will eventually be farther from the origin.

4-2-2a

73. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

A particle moves along the x -axis so that its velocity at time t is given by $v(t) = 2t \cdot \cos\left(\frac{t^3}{4}\right)$. The particle is at position $x = 3$ at time $t = 0$.

- For $0 < t < \pi$, find all times t during which the particle is moving to the left. Give a reason for your answer.
- Find the total distance traveled by the particle from time $t = 0$ to time $t = 2$. Show the setup for your calculations.
- Find the acceleration of the particle at time $t = 2$. Show the setup for your calculations. Is the particle speeding up or slowing down at time $t = 2$? Give a reason for your answer.
- Find the rightmost position of the particle for $0 \leq t \leq \pi$. Justify your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response that considers the sign of $v(t)$ earns the first point.

A response that identifies $t = 1.845$ (or $\sqrt[3]{2\pi}$) and/or $t = 2.661$ (or $\sqrt[3]{6\pi}$) and no other values in the interval $0 < t < \pi$ earns the first point.

A response of only $1.845 < t < 2.661$ earns the first point but not the second point.



0	1	2
---	---	---

4-2-2a

The student response accurately includes **both** of the criteria below.

☐ Considers $v(t) = 0$

☐ Answer with reason

Solution:

The particle is moving to the left when $v(t) < 0$.

$$0 < t < \pi \text{ and } v(t) = 0$$

$$\Rightarrow t = A = 1.845270 \text{ and } t = B = 2.661340$$

The particle is moving left for $1.845 < t < 2.661$ because $v(t) < 0$ in this interval.

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned only for $\int_0^2 |v(t)| dt$ or $\int_0^{1.845} v(t) dt - \int_{1.845}^2 v(t) dt$ or an equivalent expression, with or without the differential.

The second point is earned for an answer of 2.595 regardless of whether the first point was earned.

Degree mode: A response that presents answers obtained by using a calculator in degree mode does not earn the first point it would have otherwise earned. The response is eligible for all subsequent points (unless no answer is possible in degree mode or the question is made simpler by using degree mode). In degree mode,

$$\int_0^2 |v(t)| dt = 3.999391.$$



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Integral

☐ Answer

Solution:

$$\int_0^2 |v(t)| dt = 2.595422$$

4-2-2a

The total distance traveled by the particle from time $t = 0$ to time $t = 2$ is 2.595.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The minimum work needed to earn the first point is $v'(2) = -11.744$ (or -11.743).

$a(2) = -11.743863$ is not sufficient to earn the first point. The connection $v'(t) = a(t)$ or $v'(2) = a(2)$ must be explicitly made.

A response that writes an incorrect statement such as $v'(t) = a(2)$ or $dv = a = -11.744$ does not earn the first point but is eligible for the second point. Note that a response that presents $v' = a = -11.744$ at $t = 2$ does earn the first point.

A response does not need to find the value of $v(2)$; the sign is sufficient. However, a response that reports an incorrect sign or value of $v(2)$ does not earn the second point.

If a value for $v(2)$ is presented, that value must be accurate to the number of decimals presented, for zero up to three decimal places.

A response must declare a value for $a(2)$ to be eligible to earn the second point. In order to earn the second point, the presented conclusion must be consistent with a negative velocity at time $t = 2$ and the presented value of $a(2)$.

Degree mode: In degree mode, $v'(2) = 1.991472$ and $v(2) = 3.997563$. A response that declares $v'(2) = 1.991472$ will earn the second point in the presence of $v(2) = 3.997563$ or $v(2) > 0$ and the conclusion that the particle is speeding up.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Acceleration with setup
- ☐ Speeding up with reason

Solution:

$$v'(2) = a(2) = -11.743863$$

The acceleration at time $t = 2$ is -11.744 (or -11.743).

$$v(2) = -1.664587 < 0$$

4-2-2a

The particle is speeding up at time $t = 2$ because the particle's velocity and acceleration are both negative at that time.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for an attempt to evaluate $x(A)$.

The second point is earned for considering both 0 and π as candidates, verbally or in a table.

To earn the third point the response must identify the rightmost position at 5.471 and either:

- evaluate x at $t = 0$, $t = A$, $t = B$, and $t = \pi$, or
- evaluate x at $t = A$ and $t = \pi$ and eliminate $t = 0$ and $t = B$ from consideration by arguing that $0 < t < A \Rightarrow v(t) > 0$ and $B < t < \pi \Rightarrow v(t) > 0$.

All evaluations must be correct for the number of digits presented for one up to three decimal places.

Degree mode: The response is only eligible to earn the first two points.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Considers $x(A)$
- ☐ Considers both endpoints
- ☐ Answer with justification

Solution:

$$0 \leq t \leq \pi \text{ and } v(t) = 0$$

$$\Rightarrow t = 0, t = A = 1.845270, \text{ and } t = B = 2.661340$$

Because $v(t)$ is continuous on $[0, \pi]$, the candidates for the time at which the particle is at its rightmost position are $t = 0$, $t = A$, $t = B$, and $t = \pi$.

$$x(0) = 3$$

$$x(A) = 3 + \int_0^A v(t) dt = 5.471352$$

4-2-2a

$$x(B) = 3 + \int_0^B v(t) \, dt = 3.151377$$

$$x(\pi) = 3 + \int_0^\pi v(t) \, dt = 4.972567$$

The rightmost position of the particle for $0 \leq t \leq \pi$ is $x = 5.471$.

74. A particle moves along the x-axis so that at any time $t \geq 0$ its position is given by $x(t) = t^3 - 3t^2 - 9t + 1$. For what values of t is the particle at rest?

- (A) No values
(B) 1 only
(C) 3 only
(D) 5 only
(E) 1 and 3



75. A particle moves along the x-axis so that at time $t > 0$ its position is given by $x(t) = 12e^{-t} \sin t$. What is the first time t at which the velocity of the particle is zero?

- (A) $\frac{\pi}{4}$
(B) $\frac{\pi}{2}$
(C) $\frac{3\pi}{4}$
(D) π



4-2-2a

A particle moves on the x -axis so that its position at any time $t \geq 0$ is given by $x(t) = 2te^{-t}$.

76. Find the velocity of the particle when its acceleration is 0.

Part B

1 point is earned for correctly solving $x''(t) = 0$

$$x''(t) = -2e^{-t}(2 - t) = 0$$

$$t = 2$$

1 point is earned for correctly computing $x'(t_0)$ for all solutions found

$$x'(2) = v(2) = 2e^{-2}(1 - 2)$$

$$= \frac{-2}{e^2}$$

$$\approx -0.271$$



0	1	2
---	---	---

The student response earns two of the following points:

1 point is earned for correctly solving $x''(t) = 0$

$$x''(t) = -2e^{-t}(2 - t) = 0$$

$$t = 2$$

1 point is earned for correctly computing $x'(t_0)$ for all solutions found

$$x'(2) = v(2) = 2e^{-2}(1 - 2)$$

$$= \frac{-2}{e^2}$$

$$\approx -0.271$$

4-2-2a

77. Find the acceleration of the particle at $t=0$

Part A

1 point is earned for correctly finding $x'(t)$

$$v(t) = x'(t) = 2e^{-t} - 2te^{-t}$$

1 point is earned for correctly finding $x''(t)$

1 point is earned for correctly identifying $x''(t)$ with acceleration

$$a(t) = x''(t) = -2e^{-t} - 2e^{-t} + 2te^{-t}$$

$$a(0) = x''(0) = -2 - 2 = -4$$



0	1	2	3	4
---	---	---	---	---

The student response earns four of the following points:

1 point is earned for correctly finding $x'(t)$

$$v(t) = x'(t) = 2e^{-t} - 2te^{-t}$$

1 point is earned for correctly finding $x''(t)$

1 point is earned for correctly identifying $x''(t)$ with acceleration

$$a(t) = x''(t) = -2e^{-t} - 2e^{-t} + 2te^{-t}$$

1 point is earned for correctly computing $x''(0)$

$$a(0) = x''(0) = -2 - 2 = -4$$

4-2-2a

78. A particle moves on the x -axis so that its position at any time $t \geq 0$ is given by $x(t) = 2te^{-t}$.
- Find the acceleration of the particle at $t = 0$.
 - Find the velocity of the particle when its acceleration is 0.
 - Find the total distance traveled by the particle from $t = 0$ to $t = 5$.

Part A

1 point is earned for correctly finding $x'(t)$

1 point is earned for correctly finding $x''(t)$

1 point is earned for correctly identifying $x''(t)$ with acceleration

1 point is earned for compute $x''(0)$

$$v(t) = x'(t) = 2e^{-t} - 2te^{-t}$$

$$a(t) = x''(t) = -2e^{-t} - 2e^{-t} + 2te^{-t}$$

$$a(0) = x''(0) = -2 - 2 = -4$$



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for correctly finding $x'(t)$

1 point is earned for correctly finding $x''(t)$

1 point is earned for correctly identifying $x''(t)$ with acceleration

1 point is earned for compute $x''(0)$

$$v(t) = x'(t) = 2e^{-t} - 2te^{-t}$$

$$a(t) = x''(t) = -2e^{-t} - 2e^{-t} + 2te^{-t}$$

$$a(0) = x''(0) = -2 - 2 = -4$$

Part B

1 point is earned for correctly solving $x''(t) = 0$

1 point is earned for correctly computing $x'(t_0)$ for all solutions found

4-2-2a

$$x''(t) = -2e^{-t}(2 - t) = 0$$

$$t = 2$$

$$x'(2) = v(2) = 2e^{-2}(1 - 2)$$

$$= \frac{-2}{e^2}$$

$$\approx -0.271$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for correctly solving $x''(t) = 0$

1 point is earned for correctly computing $x'(t_0)$ for all solutions found

$$x''(t) = -2e^{-t}(2 - t) = 0$$

$$t = 2$$

$$x'(2) = v(2) = 2e^{-2}(1 - 2)$$

$$= \frac{-2}{e^2}$$

$$\approx -0.271$$

Part C

1 point is earned for correctly solving $x'(t) = 0$ or otherwise identifying turning point

1 point is earned for correctly evaluating $x(0)$, $x(5)$, and $x(\text{turning point})$

1 point is earned for the correct answer

Note: Max 1/3 if no turning point or turning point ignored

$$x'(t) = v(t) = 2e^{-t}(1 - t) = 0$$

$$t = 1$$

$$x(0) = 0$$

$$x(1) = \frac{2}{e} \approx 0.736$$

$$x(5) = \frac{10}{e^5} \approx 0.067$$

4-2-2a

$$\text{Distance} = \frac{2}{e} + \frac{2}{e} - \frac{10}{e^5}$$

$$\approx 1.404$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for correctly solving $x'(t) = 0$ or otherwise identifying turning point

1 point is earned for correctly evaluating $x(0)$, $x(5)$, and $x(\text{turning point})$

1 point is earned for the correct answer

Note: Max 1/3 if no turning point or turning point ignored

$$x'(t) = v(t) = 2e^{-t}(1 - t) = 0$$

$$t = 1$$

$$x(0) = 0$$

$$x(1) = \frac{2}{e} \approx 0.736$$

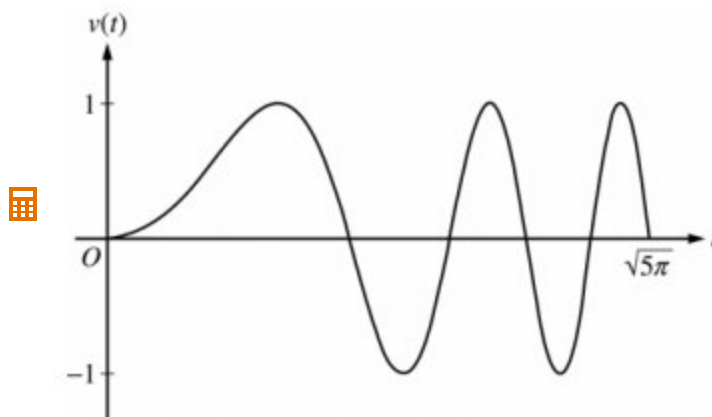
$$x(5) = \frac{10}{e^5} \approx 0.067$$

$$\text{Distance} = \frac{2}{e} + \frac{2}{e} - \frac{10}{e^5}$$

$$\approx 1.404$$

4-2-2a

79.



A particle moves along the x -axis so that its velocity v at time $t \geq 0$ is given by $v(t) = \sin(t^2)$. The graph of v is shown above for $0 \leq t \leq \sqrt{5\pi}$. The position of the particle at time t is $x(t)$ and its position at time $t = 0$ is $x(0) = 5$.

- Find the acceleration of the particle at time $t = 3$.
- Find the total distance traveled by the particle from time $t = 0$ to $t = 3$.
- Find the position of the particle at time $t = 3$.
- For $0 \leq t \leq \sqrt{5\pi}$, find the time t at which the particle is farthest to the right. Explain your answer.

Part A

1 point is earned for $a(3)$

$$a(3) = v'(3) = 6 \cos 9 = -5.466 \text{ or } -5.467$$



0	1
---	---

The student response earns all of the following points:

1 point is earned for $a(3)$

$$a(3) = v'(3) = 6 \cos 9 = -5.466 \text{ or } -5.467$$

Part B

1 point is earned for the setup

1 point is earned for the answer

$$\text{Distance} = \int_0^3 |v(t)| dt = 1.702$$

4-2-2a

OR

For $0 < t < 3$, $v(t) = 0$ when $t = \sqrt{\pi} = 1.77245$ and $t = \sqrt{2\pi} = 2.50663$

$$x(0) = 5$$

$$x(\sqrt{\pi}) = 5 + \int_0^{\sqrt{\pi}} v(t) dt = 5.89483$$

$$x(\sqrt{2\pi}) = 5 + \int_0^{\sqrt{2\pi}} v(t) dt = 5.43041$$

$$x(3) = 5 + \int_0^3 v(t) dt = 5.77356$$

$$|x(\sqrt{\pi}) - x(0)| + |x(\sqrt{2\pi}) - x(\sqrt{\pi})| + |x(3) - x(\sqrt{2\pi})| = 1.702$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the setup

1 point is earned for the answer

$$\text{Distance} = \int_0^3 |v(t)| dt = 1.702$$

OR

For $0 < t < 3$, $v(t) = 0$ when $t = \sqrt{\pi} = 1.77245$ and $t = \sqrt{2\pi} = 2.50663$

$$x(0) = 5$$

4-2-2a

$$x(\sqrt{\pi}) = 5 + \int_0^{\sqrt{\pi}} v(t) dt = 5.89483$$

$$x(\sqrt{2\pi}) = 5 + \int_0^{\sqrt{2\pi}} v(t) dt = 5.43041$$

$$x(3) = 5 + \int_0^3 v(t) dt = 5.77356$$

$$|x(\sqrt{\pi}) - x(0)| + |x(\sqrt{2\pi}) - x(\sqrt{\pi})| + |x(3) - x(\sqrt{2\pi})| = 1.702$$

Part C

1 point is earned for integrand

1 point is earned if uses $x(0) = 5$

1 point is earned for the answer

$$x(3) = 5 + \int_0^3 v(t) dt = 5.773 \text{ or } 5.774$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for integrand

1 point is earned if uses $x(0) = 5$

1 point is earned for the answer

$$x(3) = 5 + \int_0^3 v(t) dt = 5.773 \text{ or } 5.774$$

Part D

1 point is earned if sets $v(t) = 0$

1 point is earned for the answer

1 point is earned for the reason

4-2-2a

The particle's rightmost position occurs at time $t = \sqrt{\pi} = 1.772$.

The particle changes from moving right to moving left at those times t for which $v(t) = 0$ with $v(t)$ changing from positive to negative, namely at $t = \sqrt{\pi}, \sqrt{3\pi}, \sqrt{5\pi}$ ($t = 1.772, 3.070, 3.963$).

Using $x(T) = 5 + \int_0^T v(t)dt$, the particle's positions at the times it changes from rightward to leftward movement are:

$$\begin{array}{l} T : 0 \quad \sqrt{\pi} \quad \sqrt{3\pi} \quad \sqrt{5\pi} \\ x(T) : 5 \quad 5.895 \quad 5.788 \quad 5.752 \end{array}$$

The particle is farthest to the right when $T = \sqrt{\pi}$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned if sets $v(t) = 0$

1 point is earned for the answer

1 point is earned for the reason

The particle's rightmost position occurs at time $t = \sqrt{\pi} = 1.772$.

The particle changes from moving right to moving left at those times t for which $v(t) = 0$ with $v(t)$ changing from positive to negative, namely at $t = \sqrt{\pi}, \sqrt{3\pi}, \sqrt{5\pi}$ ($t = 1.772, 3.070, 3.963$).

Using $x(T) = 5 + \int_0^T v(t)dt$, the particle's positions at the times it changes from rightward to leftward movement are:

4-2-2a

$$\begin{array}{l} T : 0 \quad \sqrt{\pi} \quad \sqrt{3\pi} \quad \sqrt{5\pi} \\ x(T) : 5 \quad 5.895 \quad 5.788 \quad 5.752 \end{array}$$

The particle is farthest to the right when $T = \sqrt{\pi}$.

4-2-2a

For $0 \leq t \leq 6$, a particle is moving along the x -axis. The particle's position, $x(t)$, is not explicitly given. The velocity of the particle is given by $v(t) = 2\sin(e^{t/4}) + 1$. The acceleration of the particle is given by $a(t) = \frac{1}{2}e^{t/4}\cos(e^{t/4})$ and $x(0) = 2$.

80.  For $0 \leq t \leq 6$ the particle changes direction exactly once. Find the position of the particle at that time.

Part D

1 point is earned for considering $v(t) = 0$

1 point is earned for integral

1 point is earned for the answer

$v(t) = 0$ when $t = 5.19552$. Let $b = 5.19552$

$v(t)$ changes sign from positive to negative at time $t = b$.

$$x(b) = 2 + \int_0^b v(t) dt = 14.134 \text{ or } 14.135$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for considering $v(t) = 0$

1 point is earned for integral

1 point is earned for the answer

$v(t) = 0$ when $t = 5.19552$. Let $b = 5.19552$

$v(t)$ changes sign from positive to negative at time $t = b$.

$$x(b) = 2 + \int_0^b v(t) dt = 14.134 \text{ or } 14.135$$

4-2-2a

81.  Is the speed of the particle increasing or decreasing at time $t = 5.5$? Give a reason for your answer.
-

Part A

2 point is earned for conclusion with reason

$$v(5.5) = -0.45337, a(5.5) = -1.35851$$

The speed is increasing at time $t = 5.5$, because velocity and acceleration have the same sign.



0	1	2
---	---	---

The student response earns all of the following points:

2 point is earned for conclusion with reason

$$v(5.5) = -0.45337, a(5.5) = -1.35851$$

The speed is increasing at time $t = 5.5$, because velocity and acceleration have the same sign.

4-2-2a

82.  For $0 \leq t \leq 6$, a particle is moving along the x -axis. The particle's position, $x(t)$, is not explicitly given. The velocity of the particle is given by $v(t) = 2\sin(e^{t/4}) + 1$. The acceleration of the particle is given by $a(t) = 1/2e^{t/4}\cos(e^{t/4})$ and $x(0) = 2$.

- (a) Is the speed of the particle increasing or decreasing at time $t = 5.5$? Give a reason for your answer.
 (b) Find the average velocity of the particle for the time period $0 \leq t \leq 6$.
 (c) Find the total distance traveled by the particle from time $t = 0$ to $t = 6$.
 (d) For $0 \leq t \leq 6$, the particle changes direction exactly once. Find the position of the particle at that time.

Part A

2 points are earned for conclusion with reason

$$v(5.5) = -0.45337, a(5.5) = -1.35851$$

The speed is increasing at time $t = 5.5$, because velocity and acceleration have the same sign.



0	1	2
---	---	---

The student response earns all of the following points:

2 points are earned for conclusion with reason

$$v(5.5) = -0.45337, a(5.5) = -1.35851$$

The speed is increasing at time $t = 5.5$, because velocity and acceleration have the same sign.

Part B

1 point is earned for integral

1 point is earned for the answer

$$\text{Average velocity} = \frac{1}{6} \int_0^6 v(t) dt = 1.949$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for integral

4-2-2a

1 point is earned for the answer

$$\text{Average velocity} = \frac{1}{6} \int_0^6 v(t) dt = 1.949$$

Part C

1 point is earned for integral

1 point is earned for the answer

$$\text{Distance} = \int_0^6 |v(t)| dt = 12.573$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for integral

1 point is earned for the answer

$$\text{Distance} = \int_0^6 |v(t)| dt = 12.573$$

Part D

1 point is earned for considering $v(t) = 0$

1 point is earned for integral

1 point is earned for the answer

$v(t) = 0$ when $t = 5.19552$. Let $b = 5.19552$.

$v(t)$ changes sign from positive to negative at time $t = b$.

$$x(b) = 2 + \int_0^b v(t) dt = 14.134 \text{ or } 14.135$$



0	1	2	3
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4-2-2a

The student response earns all of the following points:

1 point is earned for considering $v(t) = 0$

1 point is earned for integral

1 point is earned for the answer

$v(t) = 0$ when $t = 5.19552$. Let $b = 5.19552$.

$v(t)$ changes sign from positive to negative at time $t = b$.

$$x(b) = 2 + \int_0^b v(t) dt = 14.134 \text{ or } 14.135$$

4-2-2a

For $0 \leq t \leq 12$, a particle moves along the x -axis. The velocity of the particle at time t is given by $v(t) = \cos(\pi/6t)$. The particle is at position $x = -2$ at time $t = 0$.

83. Find the acceleration of the particle at time t . Is the speed of the particle increasing, decreasing, or neither at time $t = 4$? Explain your reasoning.

Part C

1 point is earned for $a(t)$

2 points are earned for conclusion with reason

$$a(t) = -\frac{\pi}{6}\sin\left(\frac{\pi}{6}t\right)$$

$$a(4) = -\frac{\pi}{6}\sin\left(\frac{2\pi}{3}\right) = -\frac{\sqrt{3}\pi}{12} < 0$$

$$v(4) = \cos\left(\frac{2\pi}{3}\right) = -\frac{1}{2} < 0$$

The speed is increasing at time $t = 4$, because velocity and acceleration have the same sign.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for $a(t)$

2 points are earned for conclusion with reason

$$a(t) = -\frac{\pi}{6}\sin\left(\frac{\pi}{6}t\right)$$

$$a(4) = -\frac{\pi}{6}\sin\left(\frac{2\pi}{3}\right) = -\frac{\sqrt{3}\pi}{12} < 0$$

$$v(4) = \cos\left(\frac{2\pi}{3}\right) = -\frac{1}{2} < 0$$

The speed is increasing at time $t = 4$, because velocity and acceleration have the same sign.

4-2-2a

84. For $0 \leq t \leq 12$, when is the particle moving to the left?

Part A

1 point is earned for considering $v(t)=0$

1 point is earned for the interval

$$v(t) = \cos\left(\frac{\pi}{6}t\right) = 0 \Rightarrow t = 3, 9$$

The particle is moving to the left when $v(t) < 0$.

This occurs when $3 < t < 9$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for considering $v(t)=0$

1 point is earned for the interval

$$v(t) = \cos\left(\frac{\pi}{6}t\right) = 0 \Rightarrow t = 3, 9$$

The particle is moving to the left when $v(t) < 0$.

This occurs when $3 < t < 9$

4-2-2a

85. For $0 \leq t \leq 12$, a particle moves along the x -axis. The velocity of the particle at time t is given by $v(t) = \cos\left(\frac{\pi}{6}t\right)$. The particle is at position $x = -2$ at time $t = 0$.
- (a) For $0 \leq t \leq 12$, when is the particle moving to the left?
- (b) Write, but do not evaluate, an integral expression that gives the total distance traveled by the particle from time $t = 0$ to time $t = 6$.
- (c) Find the acceleration of the particle at time t . Is the speed of the particle increasing, decreasing, or neither at time $t = 4$? Explain your reasoning.
- (d) Find the position of the particle at time $t = 4$.

Part A

1 point is earned for considering $v(t) = 0$

1 point is earned for the interval

$$v(t) = \cos\left(\frac{\pi}{6}t\right) = 0 \Rightarrow t = 3, 9$$

The particle is moving to the left when $v(t) < 0$.

This occurs when $3 < t < 9$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for considering $v(t) = 0$

1 point is earned for the interval

$$v(t) = \cos\left(\frac{\pi}{6}t\right) = 0 \Rightarrow t = 3, 9$$

The particle is moving to the left when $v(t) < 0$.

This occurs when $3 < t < 9$.

Part B

1 point is earned for the answer

$$\int_0^6 |v(t)| dt$$

4-2-2a



0	1
---	---

The student response earns all of the following points:

1 point is earned for the answer

$$\int_0^6 |v(t)| dt$$

Part C

1 point is earned for $a(t)$

2 points are earned for conclusion with reason

$$a(t) = -\frac{\pi}{6} \sin\left(\frac{\pi}{6}t\right)$$

$$a(4) = -\frac{\pi}{6} \sin\left(\frac{2\pi}{3}\right) = -\frac{\sqrt{3}\pi}{12} < 0$$

$$v(4) = \cos\left(\frac{2\pi}{3}\right) = -\frac{1}{2} < 0$$

The speed is increasing at time $t = 4$, because velocity and acceleration have the same sign.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for $a(t)$

2 points are earned for conclusion with reason

$$a(t) = -\frac{\pi}{6} \sin\left(\frac{\pi}{6}t\right)$$

$$a(4) = -\frac{\pi}{6} \sin\left(\frac{2\pi}{3}\right) = -\frac{\sqrt{3}\pi}{12} < 0$$

$$v(4) = \cos\left(\frac{2\pi}{3}\right) = -\frac{1}{2} < 0$$

The speed is increasing at time $t = 4$, because velocity and acceleration have the same sign.

4-2-2a

Part D

1 point is earned for the antiderivative

1 point is earned for using initial condition

1 point is earned for the answer

$$\begin{aligned}
 x(4) &= -2 + \int_0^4 \cos\left(\frac{\pi}{6}t\right) dt \\
 &= -2 + \left[\frac{6}{\pi} \sin\left(\frac{\pi}{6}t\right)\right]_0^4 \\
 &= -2 + \frac{6}{\pi} \left[\sin\left(\frac{2\pi}{3}\right) - 0\right] \\
 &= -2 + \frac{6}{\pi} \cdot \frac{\sqrt{3}}{2} = -2 + \frac{3\sqrt{3}}{\pi}
 \end{aligned}$$



0	1	2	3
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The student response earns all of the following points:

1 point is earned for the antiderivative

1 point is earned for using initial condition

1 point is earned for the answer

$$\begin{aligned}
 x(4) &= -2 + \int_0^4 \cos\left(\frac{\pi}{6}t\right) dt \\
 &= -2 + \left[\frac{6}{\pi} \sin\left(\frac{\pi}{6}t\right)\right]_0^4 \\
 &= -2 + \frac{6}{\pi} \left[\sin\left(\frac{2\pi}{3}\right) - 0\right] \\
 &= -2 + \frac{6}{\pi} \cdot \frac{\sqrt{3}}{2} = -2 + \frac{3\sqrt{3}}{\pi}
 \end{aligned}$$

4-2-2a

A particle moves along a straight line. For $0 \leq t \leq 5$, the velocity of the particle is given by $v(t) = -2 + (t^2 + 3t)^{6/5} - t^3$, and the position of the particle is given by $s(t)$. It is known that $s(0) = 10$.

86.  Is the speed of the particle increasing or decreasing at time $t=4$? Give a reason for your answer.

Part D

2 points are earned for the conclusion with reason

$$v(4) = -11.475758 < 0, a(4) = v'(4) = -22.295714 < 0$$

The speed is increasing at time $t=4$ because velocity and acceleration have the same sign.



0	1	2
---	---	---

The student response earns all of the following points:

2 point is earned for the conclusion with reason

$$v(4) = -11.475758 < 0, a(4) = v'(4) = -22.295714 < 0$$

The speed is increasing at time $t=4$ because velocity and acceleration have the same sign.

4-2-2a

87.  Find all times t in the interval $0 \leq t \leq 5$ at which the particle changes direction. Justify your answer.

Part C

1 point is earned for considering $v(t)=0$

2 points are earned for answering with justification

$v(t)=0$ when $t=0.536033, 3.317756$ $v(t)$ changes sign from negative to positive at time $t=0.536033$.

$v(t)$ changes sign from positive to negative at time $t=3.317756$. Therefore, the particle changes direction at time $t=0.536$ and time $t=3.318$ (or 3.317).



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for considering $v(t)=0$

2 points are earned for answering with justification

$v(t)=0$ when $t=0.536033, 3.317756$

$v(t)$ changes sign from negative to positive at time $t=0.536033$.

$v(t)$ changes sign from positive to negative at time $t=3.317756$.

Therefore, the particle changes direction at time $t=0.536$ and time $t=3.318$ (or 3.317).

4-2-2a

88.  Find all values of t in the interval $2 \leq t \leq 4$ for which the speed of the particle is 2.
-

Part A

1 point is earned for considering $|v(t)|=2$

1 point is earned for the answer

Solve $|v(t)|=2$ on $2 \leq t \leq 4$.

$t=3.128$ (or 3.127) and $t=3.473$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for considering $|v(t)|=2$

1 point is earned for the answer

Solve $|v(t)|=2$ on $2 \leq t \leq 4$.

$t=3.128$ (or 3.127) and $t=3.473$

4-2-2a

89.  A particle moves along a straight line. For $0 \leq t \leq 5$, the velocity of the particle is given by $v(t) = -2 + (t^2 + 3t)^{6/5} - t^3$, and the position of the particle is given by $s(t)$. It is known that $s(0) = 10$.

- (a) Find all values of t in the interval $2 \leq t \leq 4$ for which the speed of the particle is 2.
- (b) Write an expression involving an integral that gives the position $s(t)$. Use this expression to find the position of the particle at time $t = 5$.
- (c) Find all times t in the interval $0 \leq t \leq 5$ at which the particle changes direction. Justify your answer.
- (d) Is the speed of the particle increasing or decreasing at time $t = 4$? Give a reason for your answer.

Part A

1 point is earned for considering $|v(t)| = 2$

1 point is earned for the answer

Solve $|v(t)| = 2$ on $2 \leq t \leq 4$.

$t = 3.128$ (or 3.127) and $t = 3.473$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for considering $|v(t)| = 2$

1 point is earned for the answer

Solve $|v(t)| = 2$ on $2 \leq t \leq 4$.

$t = 3.128$ (or 3.127) and $t = 3.473$

Part B

1 point is earned for $s(t)$

1 point is earned for $s(5)$

$$s(t) = 10 + \int_0^t v(x) dx$$

$$s(5) = 10 + \int_0^5 v(x) dx = -9.207$$

4-2-2a



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $s(t)$

1 point is earned for $s(5)$

$$s(t) = 10 + \int_0^t v(x) dx$$

$$s(5) = 10 + \int_0^5 v(x) dx = -9.207$$

Part C

1 point is earned for considering $v(t) = 0$

2 points are earned for answering with justification

$v(t) = 0$ when $t = 0.536033, 3.317756$

$v(t)$ changes sign from negative to positive at time $t = 0.536033$.

$v(t)$ changes sign from positive to negative at time $t = 3.317756$.

Therefore, the particle changes direction at time $t = 0.536$ and time $t = 3.318$ (or 3.317).



0	1	2	3
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The student response earns all of the following points:

1 point is earned for considering $v(t) = 0$

2 points are earned for answering with justification

$v(t) = 0$ when $t = 0.536033, 3.317756$

$v(t)$ changes sign from negative to positive at time $t = 0.536033$.

$v(t)$ changes sign from positive to negative at time $t = 3.317756$.

4-2-2a

Therefore, the particle changes direction at time $t = 0.536$ and time $t = 3.318$ (or 3.317).

Part D

2 points are earned for the conclusion with reason

$$v(4) = -11.475758 < 0, a(4) = v'(4) = -22.295714 < 0$$

The speed is increasing at time $t = 4$ because velocity and acceleration have the same sign.



0	1	2
---	---	---

The student response earns all of the following points:

2 points are earned for the conclusion with reason

$$v(4) = -11.475758 < 0, a(4) = v'(4) = -22.295714 < 0$$

The speed is increasing at time $t = 4$ because velocity and acceleration have the same sign.

4-2-2a

90. Particle X moves along the positive x -axis so that its position at time $t \geq 0$ is given by $x(t) = 5t^3 - 9t^2 + 7$. Is particle X moving toward the left or toward the right at time $t = 1$? Give a reason for your answer.

Part A

The response can earn up to 2 points:

1 point: For consideration of $x'(1)$

1 point: Correct answer with reason.

$$x'(t) = 15t^2 - 18t$$

$$x'(1) = 15 - 18 = -3$$

Since $x'(1) < 0$, the particle is moving to the left at time $t = 1$



0	1	2
---	---	---

The response earns both of the following points:

1 point: For consideration of $x'(1)$

1 point: Correct answer with reason.

$$x'(t) = 15t^2 - 18t$$

$$x'(1) = 15 - 18 = -3$$

Since $x'(1) < 0$, the particle is moving to the left at time $t = 1$.